

CrossMark
click for updatesCite this: *RSC Adv.*, 2014, 4, 53442

Recent mechanistic developments and next generation catalysts for the Sonogashira coupling reaction

Milandip Karak,^a Luiz C. A. Barbosa^{*ab} and Gráinne C. Hargaden^c

The Pd-catalyzed Sonogashira reaction is a powerful method for the formation of $C_{sp^2}-C_{sp}$ bonds and has found application in a wide variety of areas including medicinal chemistry, agrochemistry, materials and electronics. Development of competent catalysts for the Sonogashira reaction is a particular scientific challenge since it is traditionally a di-metallic-mediated homogeneous catalytic process including some major drawbacks. This review provides a concise overview of the mechanistic aspects of the Cu co-catalyzed, Cu-free and Au-catalyzed Sonogashira coupling processes. More recent developments and next generation catalysts for the Sonogashira reaction are also presented. These include non transition-metal catalysts, metal free couplings and photo-induced protocols. Finally, the application of metal nanoparticles in Sonogashira reactions is presented. These include Pd nanoparticles, Pd bi- and tri-metallic nanoparticles, magnetically separable Pd/Fe₃O₄ nanoparticles, Ru nanoparticles and Au nanoparticles.

Received 22nd August 2014

Accepted 7th October 2014

DOI: 10.1039/c4ra09105a

www.rsc.org/advances

1. Introduction

Over the past 15 years, there has been a growing interest in the Sonogashira coupling reaction¹ which is one of the most powerful methods for the formation of $C_{sp^2}-C_{sp}$ bonds. The Pd catalyzed $C_{sp^2}-C_{sp}$ coupling reaction between aryl halides and terminal alkynes (Scheme 1) has become the most important method for preparing aryl alkynes and conjugated alkenynes, which are often intermediates or precursors in the synthesis of

^aDepartamento de Química, Universidade Federal de Viçosa, Av. Peter Henry Rolfs, s/n Campus Universitário, CEP 36570-900, Viçosa, MG, Brazil. E-mail: lcab@ufmg.br

^bDepartamento de Química, Universidade Federal de Minas Gerais, Av. Pres. Antônio Carlos, 6627, Campus Pampulha, CEP 31270-901, Belo Horizonte, MG, Brazil

^cSchool of Chemical and Pharmaceutical Sciences, Dublin Institute of Technology, Kevin Street, Dublin 8, Ireland



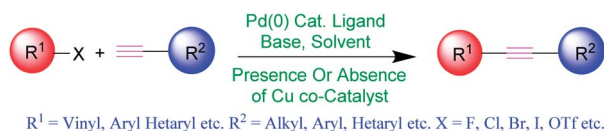
Mr Karak graduated with Honours in Chemistry in 2009 from Vidyasagar University, India, and received the MSc degree in Chemistry from Nagpur University, India in 2011. Shortly after completion of his Masters, he joined Chemgen Pharma International, India as a Research Chemist in October 2011 and spent more than one year there. In April 2013 he joined the Universidade Federal

de Viçosa (UFV) in Brazil as a doctoral student under the supervision of Professor Luiz C. A. Barbosa. His research is directed towards the design of new synthetic routes and their application to the synthesis of novel organic molecules that can lead to current pharmaceuticals, agrochemicals or new materials.



Professor Luiz C. A. Barbosa, received a BSc (1981) and MSc (1986) from Federal University of Viçosa (UFV-Brazil). In 1991 completed his PhD in Organic Synthesis at the University of Reading (UK) and spent a sabbatical year (2008–2009) in the group of Professor Timothy J. Donohoe at the University of Oxford. He started his academic career as an auxiliary professor at UFV (1983) raising to

Professor in 1997. Since 2012 he is a Full Professor at Federal University of Minas Gerais. His research interest includes the synthesis of biologically active compounds analogues of natural products. More details at <http://www.luizclaudiobarbosa.com.br>



Scheme 1 General reaction scheme for the Sonogashira reaction.

natural products,^{2,3} pharmaceuticals of biological activity,^{4,5} non-linear optical materials and molecular electronics,^{6–8} dendrimeric and polymeric materials,^{9–11} macrocycles with acetylene links¹² and polyalkynylated molecules.^{13,14}

Although the Sonogashira reaction became the most popular route for carrying out alkylation of aryl or alkenyl halides, the standard Pd catalyzed and Cu co-catalyzed Sonogashira reactions has some major drawbacks including the use of highly expensive Pd catalysts (sometimes required in high loadings), difficulties in recovering these catalysts, high reaction temperatures, air sensitivity of transition-metal complexes and the use of large amounts of ligands and amines. Another drawback of the Sonogashira reaction is the formation of homocoupled products upon exposure of the alkynes to oxidizing agents, Cu salts or air (Glaser coupling¹⁵ or Hay coupling¹⁶). The homocoupling products can be formed in considerable yields and tend to be difficult to separate from the desired products due to similar chromatographic mobility.¹⁷ With these significant drawbacks, a more efficient process is highly desirable. To date, several modifications including Pd-free,¹⁸ Cu-free,¹⁹ amine-free,²⁰ ligand-free²¹ and solvent-free²² conditions and also variations of the ligands, Pd sources, solvents and bases with a view to performing the coupling reactions more efficiently have been reported. Several modifications, including the use of inexpensive metal catalysts, such as Cu,^{23,24} Fe,²⁵ and Ni²⁶ in place of Pd in the Sonogashira reaction have been successfully described, but there has been some concerns that the observed effects were due to trace amounts of Pd contamination present in the reagents used. Even now the exact mechanism of the homogeneous Pd catalyzed and Cu co-catalyzed Sonogashira reaction is not well understood and has remained a topic of debate.



Dr Gráinne C. Hargaden received her BSc and PhD from University College Dublin. Following three years as a postdoctoral research associate at the University of Oxford in the group of Professor Tim Donohoe, she was appointed lecturer in organic chemistry at Dublin Institute of Technology in 2010. She is a principal investigator in the Centre for Research in Engineering Surface Technology (CREST), and her research

interests lie at the interface of synthetic organic chemistry, catalysis and materials science.

In view of several remarkable improvements of the Sonogashira reactions in recent years, the focus of this review is the current development of the mechanism and next generation catalysts protocols for the Sonogashira reactions with their variety of scopes and limitations.

2. Mechanistic study: a concise overview

In recent years, there has been a significant effort in the development of experimental procedures which work in the absence of Cu salts, namely, Cu-free Sonogashira reactions. Although Cu(I) plays a crucial role in the Sonogashira reaction, the addition of a Cu(I) salt under the reaction conditions leads to some drawbacks, including the induction of a Glaser¹⁵ type oxidative homocoupling of the terminal alkyne to yield the corresponding diyne.²⁷ In some cases it has been observed that Cu(I) salts have an inhibitory effect in the Sonogashira reaction.^{28,29} Several computational studies have focused on the mechanism of this Cu-free protocol.^{30–32}

2.1. Copper co-catalyzed reaction mechanism

The general Sonogashira protocol for the reaction of terminal alkynes with aryl or alkenyl halides or triflates involves a Pd(0)/Cu(I) catalytic system and at least a stoichiometric amount of base.^{33,34} The presence of a Cu(I) salt is generally believed to facilitate the transfer of the alkynyl group to the Pd catalyst by the *in situ* generation of a Cu-acetylide species and the subsequent transmetalation of this group to Pd. The generally accepted Pd-catalyzed cycle is based on four steps: (A) oxidative addition (where Pd is oxidized from Pd(0) to Pd(II)) (*cis-trans*); (B) transmetalation (Pd remains as Pd(II)); (C) *trans-cis* isomerization (Pd remains as Pd(II)); and (D) reductive elimination (where Pd is reduced from Pd(II) to Pd(0)) (Fig. 1).

Mechanistic studies of the oxidative addition step predict the formation of *cis*-Pd(II)L₂RX complexes, which have rarely been isolated.³⁵ Therefore, the isomerization of *cis*-Pd(II)L₂RX to *trans*-Pd(II)L₂RX complex is proposed (Fig. 1). This step was thoroughly investigated and four separate pathways were found (two of them autocatalytic and two solvent assisted).³⁵ Computational studies concluded that the ligand assisted mechanism was energetically the most favored.³⁶ The isomerization of the oxidative addition product *cis*-Pd(II)L₂RX to *trans*-Pd(II)L₂RX complex is usually very fast³⁷ and therefore, the *trans*-complex is generally accepted as the starting point of the deprotonation step. The next step in the Pd-catalyzed cycle would connect with the cycle of the Cu co-catalyst. Careful analysis of the catalytic cycle reveals that the terminal alkyne is not the direct nucleophile in the Pd-catalyzed cycle. It is proposed that Cu-acetylide is the nucleophile for the Pd catalyzed cycle, and is also the catalytic species for the Cu-catalyzed cycle.³⁸ Thus, a transmetalation from the Cu-acetylide formed in the Cu-cycle would generate a Pd-complex species, which gives the final coupled alkyne after *trans-cis* isomerization and reductive elimination, with regeneration of the catalyst. The reductive elimination step is usually irreversible. Recently, a report established that the

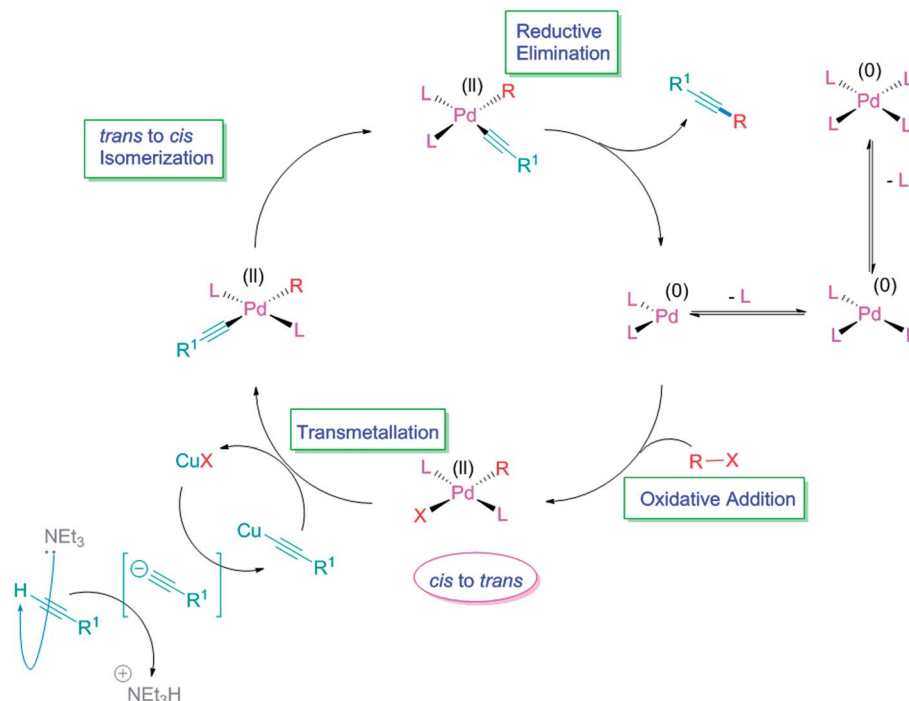


Fig. 1 Catalytic cycle for the Cu co-catalyzed Sonogashira reaction.

transmetalation step of the Sonogashira reaction is the rate-limiting step³⁹ which was demonstrated to be a Pd catalyzed and Cu co-catalyzed synergistic process and exhibits a first-order kinetic dependence on the Pd and Cu catalysts, respectively.

2.2. Copper-free reaction mechanism

Several studies have been carried out in an attempt to propose a reasonable mechanism for the Cu-free Sonogashira reaction by chemical kinetics experiments and DFT (density functional theory) calculations. The temperature dependent Sonogashira reactions for several different aryl halides, and the corresponding activation enthalpies and entropies, determined by means of Eyring plots, has established a linear correlation between $\Delta H^\ddagger$ and the Kohn–Sham HOMO energies of aryl halides. Notably, in the same year, another study attempted to explain a sophisticated mechanism for the Sonogashira reaction based on theoretical study employing DFT calculations.⁴⁰ Such studies demonstrated that there are two pathways in the oxidative addition step, the neutral and the anionic pathways, which could both give rise to *cis*-configured square planar Pd(II)–diphosphine intermediates. Initiated by this intermediate, reductive elimination of aryl acetylene is energetically favorable. More recent studies also revealed that the Pd catalyzed Cu-free Sonogashira reaction mechanism, explained by DFT calculations and theoretical mechanistic studies,^{41,42} goes *via* two most common routes: (a) carbopalladation^{43,44} and (b) deprotonation⁴⁵ (Fig. 2).

Both mechanisms share the initial oxidative addition of the organohalide R^1-X to the $Pd(0)L_2$ complex giving intermediate 1 where the metal has been oxidized from $Pd(0)$ to $Pd(II)$.

Subsequently the ligand is replaced by the alkyne, which results in the formation of complex 2, with Pd remaining as $Pd(II)$ (Fig. 2). The theoretical results demonstrated that the proposed carbopalladation mechanism has a very high energy barrier,⁴² which indicates that this mechanism does not occur under the reaction conditions. In the deprotonation mechanism,⁴⁶ the calculated Gibbs energy barriers indicate that both cationic and anionic mechanisms are feasible (Fig. 3).⁴² The cationic mechanism involves the $L-X$ ligand substitution in complex 2 (Fig. 3) giving rise to the cationic Pd complex *cis*- $[Pd(R^1)(alkyne)(L)_2]^+$ (Fig. 3; cationic pathway), which undergoes deprotonation of the alkyne by an external base and subsequent reductive elimination of $Pd(II)$ to $Pd(0)$. In contrast, in the anionic mechanism the deprotonation of the alkyne occurs first resulting in the anionic complex *cis*- $[Pd(R^1)-(acetylide)(X)(L)]^-$ (Fig. 3; anionic pathway), in which the $L-X$ ligand substitution takes place followed by the reductive elimination step to regenerate $Pd(0)$. Besides, the stabilization of a negatively charged transition state for the coupling reactions of all aryl halides in Sonogashira reactions is well described by Hammett parameter σ^- .^{47,48} Additionally, the electronic nature of the substituents directly linked to the terminal alkynes where the alkynes connected to electron withdrawing groups (EWG) favor the anionic mechanism and the alkynes bearing electron donating groups (EDG) favor the cationic pathway.⁴⁶ In addition, mechanistic studies⁴⁶ of the Cu-free version of the Sonogashira reaction have shown that the polarity and hydrogen bonding ability of the solvents are important in accelerating the reaction by stabilizing the ionic intermediates of the catalytic cycle, whereas steric bulk decreases the stabilizing ability of the solvent.⁴⁹ The ligand size influences the catalytic activity of the Pd catalyst complexes.

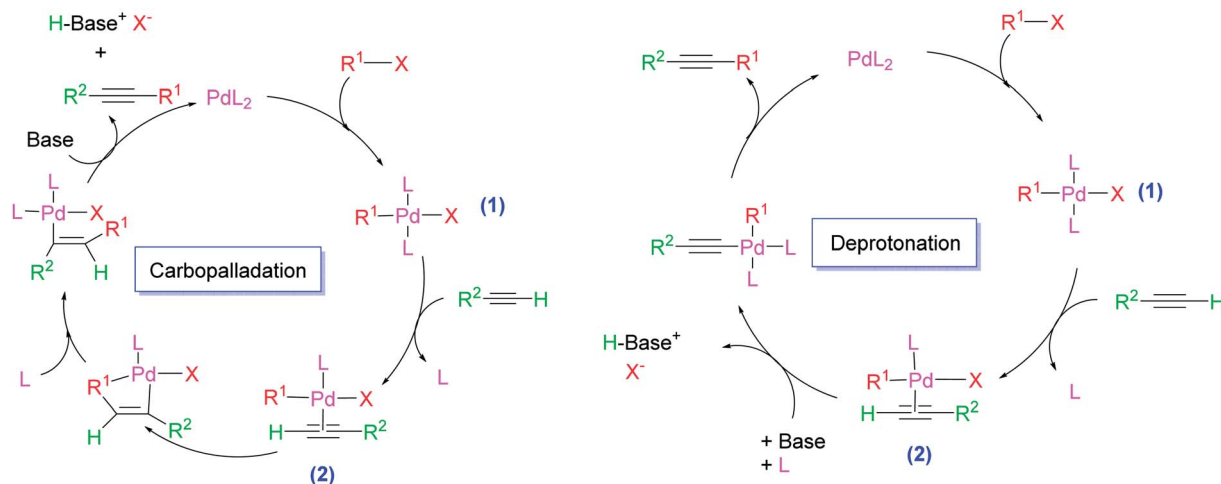


Fig. 2 Catalytic cycles proposed for the Cu-free Sonogashira reaction involving two common routes designated as carbopalladation⁴³ and deprotonation.⁴⁵

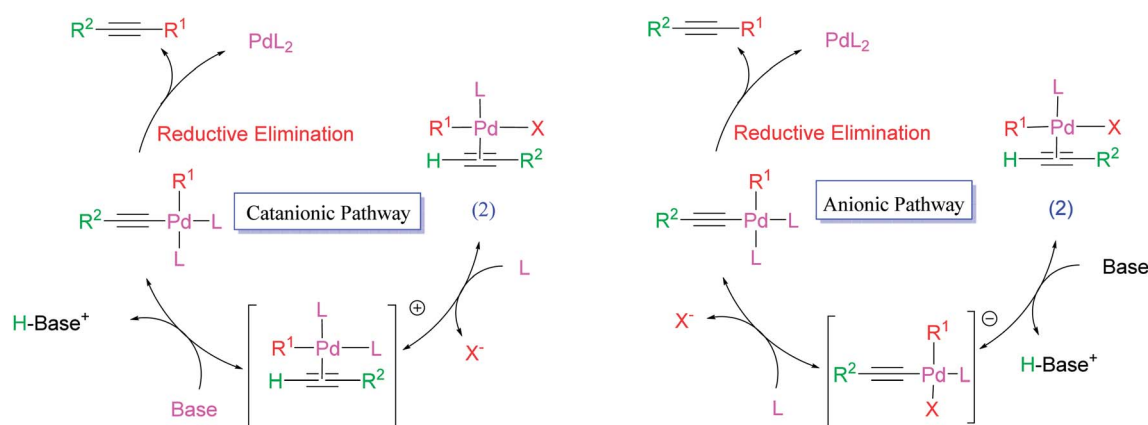


Fig. 3 Cationic and anionic pathways for the deprotonation mechanism.^{46,42}

Increasing the reactivity of the catalytic system by using bulkier phosphine ligands has been shown to be a successful method of performing the Cu-free Sonogashira reaction.⁵⁰

Recently, a study pointed out an alternative mechanism for the Sonogashira reaction named 'ionic mechanism'. This new proposal helps to explain the role of additional ligands or bases in the reaction medium.⁴² In the ionic mechanism, halide ions are always replaced by the base in the first step and results in the formation cationic Pd complex, $[\text{Pd}(\text{R}^1)(\text{base})(\text{L})_2]^+$. This mechanism simultaneously depends on the base and the acidity of the proton of the alkyne, which at the same time depends on the electron withdrawing ability of the group coordinated to the alkyne. Thus, the reaction through this mechanism is expected to be faster when alkynes bearing EWGs are used.

2.3. Gold-catalyzed reaction mechanism: a controversy

Recently, gold catalysts have excelled owing to their unique reactivity in Sonogashira reaction. Although a number of Au-catalyzed Sonogashira reactions have been reported, the mechanistic details are somewhat ambiguous. As Au(I) has the same

d^{10} configuration as Pd(0) and Cu(I), it is thought to catalyze reactions typically promoted by Pd, that is, cross-coupling reactions. From this hypothesis, research has successfully advanced to enable cross-coupling reactions with Au catalysts.

A combination of kinetic and theoretical studies⁵¹ concluded that Au is sufficiently active to promote the Sonogashira reaction. Initially, it is reasonable to consider that similar d^{10}/d^8 interplay might be operative for the two isoelectronic couples Pd(0)/Pd(II) and Au(I)/Au(III). Both d^8 metal centers are not only isoelectronic, they also form square-planar complexes. For the d^8 metal centers, Au(I) is usually coordinated in a linear manner. Although Pd(0) with common ligands is tetrahedral, it is known to dissociate in solution and it is believed that the reactive species are dicoordinated PdL_2 .⁵² However, there is an important difference in the lower oxidation state of the two systems; for PdL_2 only the oxidative addition reaction is expected, but for AuXL both oxidative addition and transmetalation reactions are plausible.^{53,54} Therefore, two cycles were considered where these two steps are transposed (Fig. 4; cycles A and B).⁵⁵ The feasibility of the two cycles and their steps

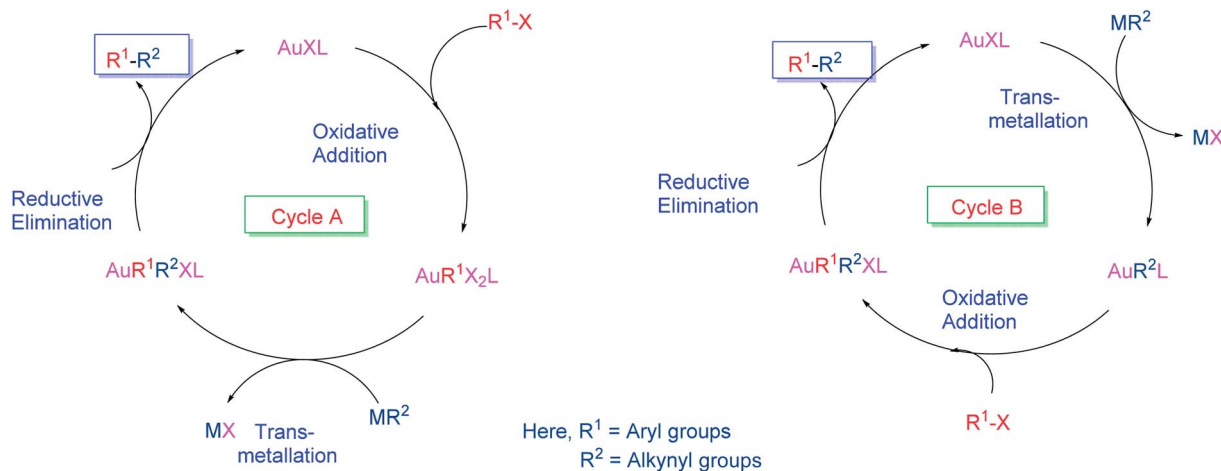


Fig. 4 Hypothetical cycles for Au-catalyzed Sonogashira reaction.^{55,56}

as isolated processes was experimentally investigated.⁵⁶ According to this experiment, Au was not capable to catalyze the Pd-free Sonogashira reaction as a sole catalyst. The investigation revealed that Au(I) is unable to activate the electrophile aryl halide by undergoing oxidative addition and the trans-metallation reaction between an aryl–Au(III) complex and other alkynyl derivative was also overruled, thus both coupling cycle frustrated. Interestingly, the addition of a catalytic amount of Pd complex results in the coupling product which suggested that the Pd catalyst performs the oxidative addition, not the Au catalyst. In these conditions the reaction would be a common Sonogashira reaction catalyzed by Pd, with Au(I) playing the role classically played by Cu(I).⁵⁷

On the other hand, Corma and co-workers^{58,51} took on the challenge to promote Au catalysts as efficient alternatives to Pd catalysts. On the basis of their hypothesis, a kinetic study of the Sonogashira reaction between iodobenzene and phenylacetylene with a series of catalyst samples consisting of a constant amount of Au/CeO₂ (99.999% Au, compared to the 99.9% Au used in the standard preparation of Au/CeO₂) and increasing amounts of Pd was carried out and concluded that Au is intrinsically active for the Sonogashira reaction between iodobenzene and phenylacetylene, and that the presence of Pd is not strictly necessary for the reaction to occur.

Over the past number of years, there has been some discussion in the literature regarding these mechanisms. For the more traditional reaction conditions the mechanism is well established, however, the actual mechanism that operates under newly developed reaction conditions remains unclear. In view of this mechanistic complexity, further studies are needed to better understand the course of Sonogashira reactions with catalysts that are practically utilized.

3. Next generation catalysts: future of the Sonogashira reaction

The original Sonogashira reaction is usually performed using a Pd–phosphine ligand complex, *e.g.* Pd(PPh₃)₄, Pd(PPh₃)₂Cl₂,

Pd(dppe)Cl₂, Pd(dppp)Cl₂, Pd(dppf)Cl₂ as the catalyst, in the presence of a catalytic amount of a Cu(I) salt as co-catalyst and an amine under homogeneous conditions. Pd(PPh₃)₄ and Pd(PPh₃)₂Cl₂ are the palladium sources most frequently employed in a Cu co-catalyzed Sonogashira reaction due to their stability and solubility.⁵⁹ In the case of deactivated bromoarenes or low reacting chloroarenes, more electron rich phosphine ligands have been employed to provide an easier oxidative insertion to aryl halides.⁶⁰ In addition, sterically ligated Pd complexes are also crucial to oxidative addition which promotes an easier dissociation from the Pd(0)L₂ resting state. In summary electron-richness favors the oxidative addition step, whereas steric bulkiness favors the formation of low coordinate and highly active Pd complexes.⁶¹ The required addition of Cu(I) halide in the standard Sonogashira reaction presents some drawbacks including environmental concerns, and the necessity to perform the coupling reaction in the absence of oxygen to minimize the occurrence of the Hay¹⁶ or Glaser¹⁵ Cu catalyzed homocoupling²⁷ of the employed terminal alkynes. In some cases, Cu has been shown to inhibit the reaction.^{28,29} Reagent grade argon or nitrogen is typically used to provide the inert atmospheric conditions required to minimize this undesired reaction, although special equipment designed to avoid the presence of oxygen, such as two-chamber reaction systems, have been developed.⁶²

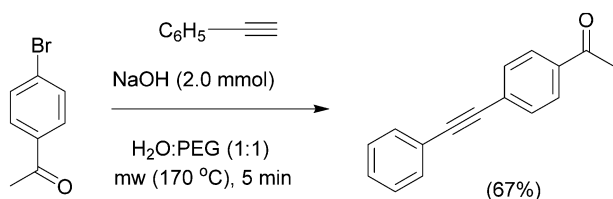
Currently the use of expensive Pd–phosphine complexes, high Pd loadings, and Pd contamination of the products has rendered Pd catalysts unpopular for some applications. This is a significant issue for large-scale industrial applications as^{27,63} the trace residues of transition metals are often difficult to remove from final products, which may be active pharmaceutical ingredients.⁶⁴ Furthermore, the presence of the Cu salt can facilitate homocoupling of the terminal alkynes and the diyne by-product may be difficult to separate from the target product.^{65,66} Consequently, it is advantageous to develop transition-metal-free cross-coupling reaction conditions. Recently, UV/visible-light photocatalysis has entered the field of cross-coupling reactions. The photochemical protocols can

easily overcome the problems associated with the conventional thermal processes, although the development of efficient methods to facilitate the recovery and reuse of metal complexes has remained an important goal. The immobilization of homogeneous/heterogeneous catalysts onto polymers,⁶⁷ carbon nanofiber,⁶⁸ clay,⁶⁹ montmorillonite,⁷⁰ silica and magnetic mesoporous silica,⁷¹ zeolite,⁷² metal oxides⁷³ and silica-starch⁷⁴ has been an expanding research area offering the advantages of easy product separation, catalyst recovery and recycling. From an economic and environmental standpoint, it is desirable to use the catalysts without employing hazardous additives as a greener alternative to conventional homogeneous or heterogeneous processes.

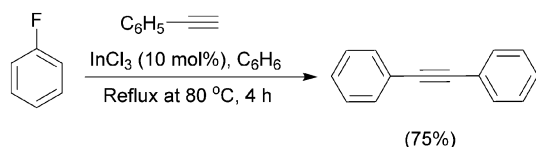
3.1. Non-transition metal based and/or metal free Sonogashira reaction

In 2003, the first example of a transition-metal free microwave assisted Sonogashira type reaction using water as the solvent, poly(ethylene glycol) (PEG) as the phase-transfer agent, and sodium hydroxide as the base was reported. The protocol worked well for a range of aryl iodides and with some aryl bromides (Scheme 2).⁷⁵ Changing the phase-transfer agent from tetrabutylammonium bromide (TBAB) to PEG had a significant effect in increasing the reaction yields.

In the same year, another study demonstrated the efficiency of microwave-assisted transition-metal free Sonogashira-type reactions⁷⁶ using water as the sole solvent, without the need for Cu(I) or any transition-metal-phosphane complex. Surprisingly, TBAB acted as the phase transfer catalyst in this protocol while PEG was not effective. In both cases inorganic bases were shown to be more efficient than organic bases. The scope of this methodology was investigated with a wide variety of well tolerated functional groups. The aryl chlorides proved to be inert under these reaction conditions, and complete regioselectivity was reported for the reaction of 4-bromo-1-chlorobenzene. In 2005, another study reported a simple transition-metal free Sonogashira-type reaction using indium-(III) trichloride as the catalyst,⁷⁷ (Scheme 3) which has been shown to promote the



Scheme 2 Transition-metal free Sonogashira reaction.



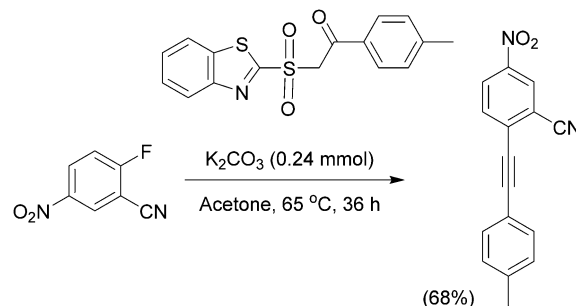
Scheme 3 Indium catalyzed Sonogashira reaction of fluorobenzene.

cross-coupling of activated and unactivated aryl iodides, bromides, chlorides, and even fluorobenzene to produce phenylacetylene, in dry benzene as the solvent.

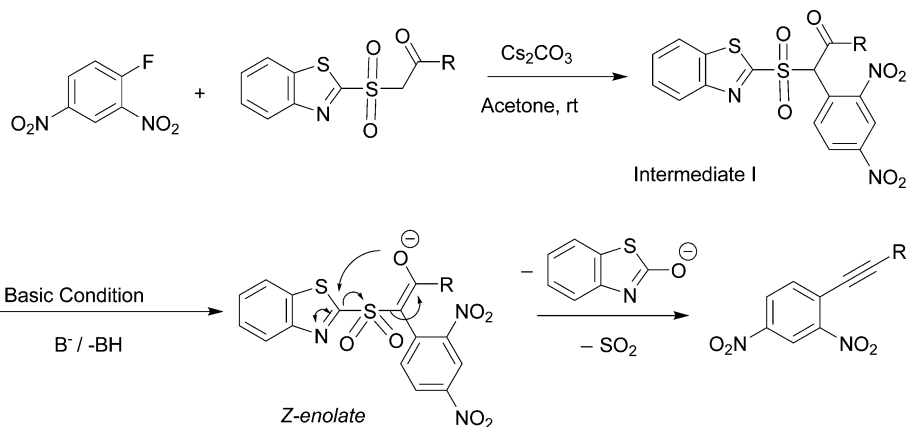
Subsequently, another transition-metal free formal Sonogashira-type reaction has been developed through nucleophilic aromatic substitution of β -carbonyl sulfones to electron-deficient aryl fluorides by using K_2CO_3 as base and acetone as the solvent (Scheme 4).⁷⁸ The scope of these reactions is demonstrated for the synthesis of Sonogashira products from a range of electron-deficient aryl fluorides with a variety of functional groups and aryl, heteroaryl, alkyl, and alkoxy substituted sulfone nucleophiles. The alkyne formation process in this formal Sonogashira reaction is affected by the steric bulk of the substituents on the nucleophile. Increased steric hindrance of the nucleophile would slow down the intermolecular S_NAr step, but speed up the intramolecular Smiles rearrangement. When the adduct is treated with base, Sonogashira product arise through α -deprotonation of the adduct where the intermediate Z -enolate subsequently rearrange and eliminate SO_2 and BTO – (BT = benzothiazol-2-yl) to yield alkynes (Scheme 5).

In 2010, another study reported a transition-metal free Sonogashira-type reaction between aryl and alkynyl Grignard reagents by using 2,2,6,6-tetramethylpiperidine- N -oxide radical (TEMPO) as a mild organic oxidant (Scheme 6).⁷⁹ Although, the practical protocol tolerates versatile functional groups, installation of substituents at the *ortho* position of the aryl-Mg derivative to suppress the undesired oxidative homocoupling reaction of aryl Grignard reagents is a major limitation of this type of oxidative cross coupling.

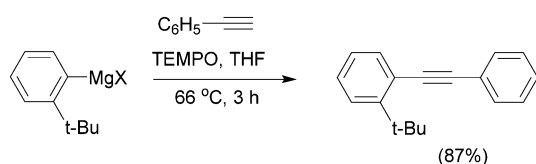
More recently, another novel Sonogashira-type reaction of terminal alkynes with unreactive aryl fluorides in the presence of sodium, sodium methoxide, and calcium hydroxide in the presence of a Grignard reagent has been developed (Scheme 7).⁸⁰ Although efficient in many cases, these protocols are limited by the narrow choice of unreactive arylfluorides and the requirement of a strong EWG in fluoroarenes for activation of the C–F bond. Mechanistic studies (Scheme 8) demonstrated electron transfer from an electron-donor species, sodium, to aryl fluoride to get free aryl radical and the fluoride anion. The fluoride anion reacts with the calcium ion to form insoluble calcium fluoride which is favorable for the cleavage of the C–F bond. The other crucial intermediate alkynylmagnesium



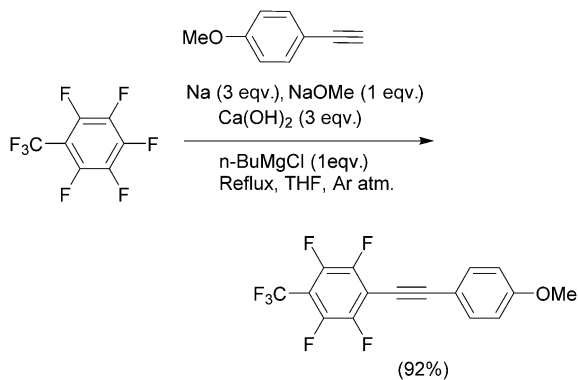
Scheme 4 One-pot alkyne formations of aromatic fluorides using β -keto sulfones.



Scheme 5 Mechanistic considerations for the formation of Sonogashira product from sulfones.



Scheme 6 Oxidative Sonogashira reaction of aryl Grignard reagent.



Scheme 7 Sonogashira reaction between phenyl acetylene and perfluorotoluene.

chloride, is generated by the reaction of aryl acetylene with butylmagnesium chloride with the assistance of sodium methoxide. In the next step, the phenyl acetylde anion attacks aryl radical in an $S_{RN}1$ manner, forming the required product and also resulting in the formation of undesired biphenyl derivatives through aryl radicals.

A metal-free Sonogashira coupling protocol between phenylacetylene and electron-deficient aryl halides catalyzed by 1,4-diazabicyclo[2.2.2]octane (DABCO) was reported.⁸¹ The reaction can be carried out under relatively mild conditions. The nucleophilicity and steric constraints of the base used are critical parameters to be considered (Scheme 9). In the proposed reaction mechanism (Scheme 10), DABCO acts as a nucleophile attacking the electrophilic carbon from the triple bond in the alkyne (Scheme 10, path a). The relatively stabilized

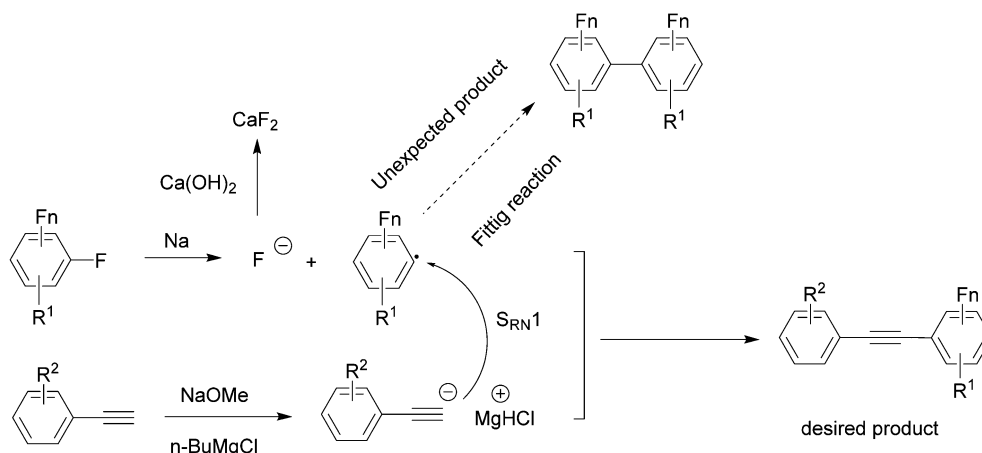
intermediate attacks the aryl halide to yield the cross-coupled product. An alternative pathway (Scheme 10, path b) involving nucleophilic substitution of the aryl halide through the iodine, followed by attack of the newly generated nucleophilic carbon in the aromatic ring to another molecule of alkyne, to give the Sonogashira coupling product. The rate determining step is much faster in path a due to nucleophilic substitution in the aryl halide being favored as the iodine has a large electronic cloud that facilitates the nucleophilic attack leaving as hydrogen iodide. EWGs in the aromatic ring reduce electron density in the aromatic ring, thus destabilizing the positive charge generated in the C–I bond and driving the attack of the nucleophile.

In another study, two sets of conditions for base-mediated, transition-metal free alkylation of aryl chlorides which proceed *via* a benzyne intermediate have been developed. In this case no mechanism has been suggested (Scheme 11).⁸² The first set of conditions involves the use of the hindered lithium 2,2,6,6-tetramethylpiperidide (TMPLi) base in a pentane–THF mixture at room temperature. The second set of conditions involves use of a metal alkoxide base in dioxane. The method also tolerates different EWGs and EDGs.

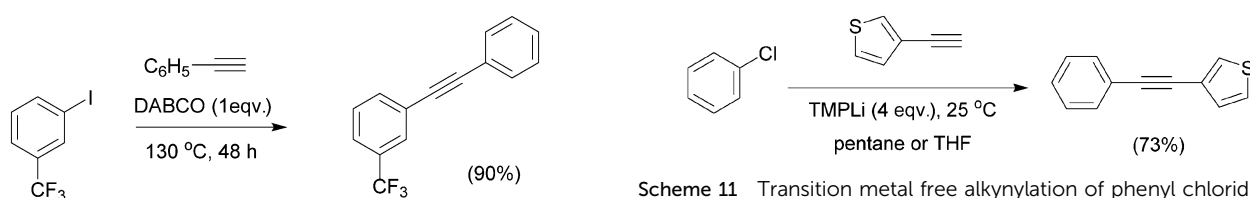
Although studies reported that only trace amounts of Pd as impurities in basic salts were able to catalyze C–C bond forming reactions, yielding the cross-coupled product in a transition-metal free reaction,⁸³ control experiments have completely ruled out the possibility of Pd contamination being responsible for the observed high yields.^{76,84}

3.2. Photo induced Sonogashira reaction

UV/Vis light-induced photocatalysis offers the possibility of initiating organic transformations with high selectivity under mild conditions. Recently, UV/Vis light-induced photocatalysis has been applied to Sonogashira reactions. A radically different approach for the synthesis of aryl acetylenes is based on the use of phenyl cations (Scheme 12). The triplet state intermediates are obtained by photolysis of aryl halides substituted with EDGs^{85,86} and addition to alkenes to form an aryl–C bond *via* a phenonium ion.^{85,87} The phenyl cations react with triple bonds



Scheme 8 Proposed reaction mechanism for the Grignard reagent assisted Sonogashira reaction.

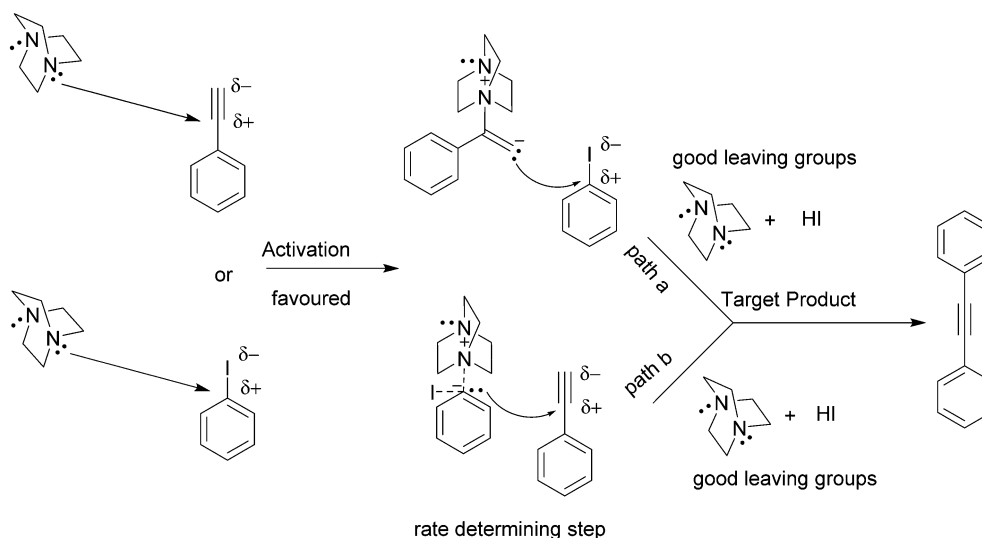


Scheme 9 Sonogashira reaction between phenyl acetylene and electron-deficient aryl halide.

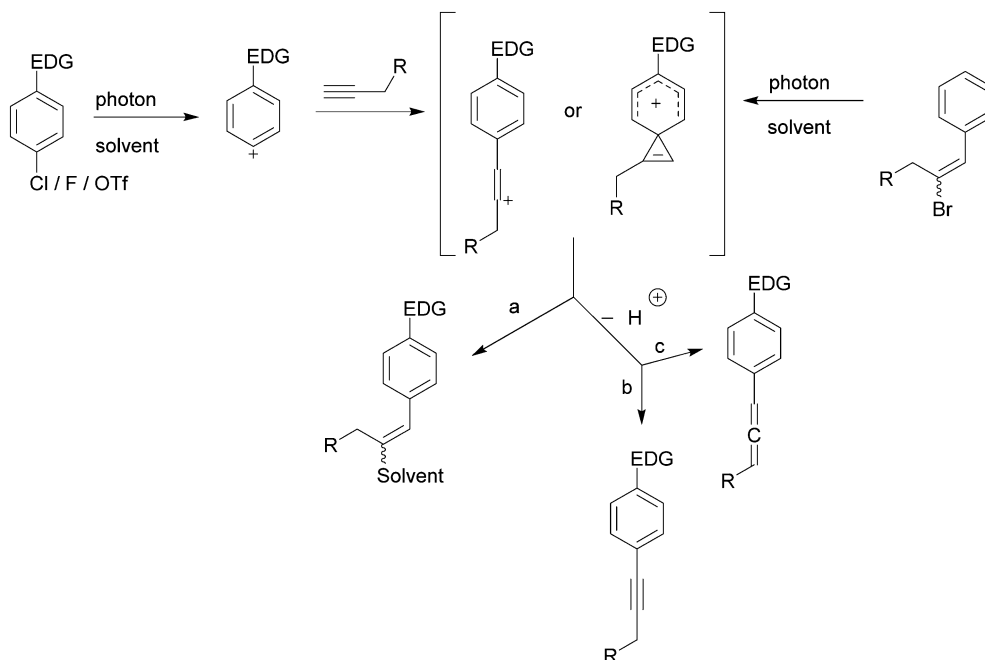
Scheme 11 Transition metal free alkylation of phenyl chloride and thienylacetylene.

to yield β -phenylvinyl cations, or more probably cyclic vinylenebenzenium ions with a nucleophilic solvent. Under these conditions, addition to form an alkoxyethyl derivative occurred (Scheme 12, path a),⁸⁸ whereas the proton elimination competed to a variable extent and led either to an aryl alkyne (Scheme 12, path b) or an aryl allene (Scheme 12, path c). Thus, the choice of reaction conditions is critical so that the desired elimination occurs to form an aryl alkyne.

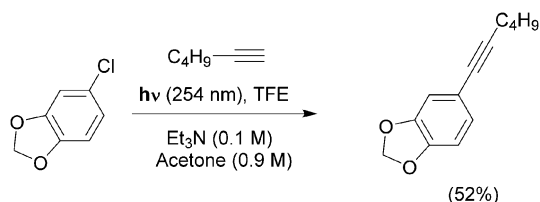
A metal-free protocol for the mild alkylation of aromatic compounds substituted with EDG has been developed which could be achieved under longer UV irradiation times at room temperature (Scheme 13).⁸⁹ In this case, UV irradiation of an aryl chloride leads to the formation of a phenyl cation, which further reacts with alkynes. As noted by the authors, this photo-initiated Sonogashira reaction has some limitations, which includes UV light absorption by solvents, aromatic alkyne reactants and products. Low reaction yields were also observed



Scheme 10 Plausible reaction pathways for the metal-free DABCO catalyzed Sonogashira reaction.



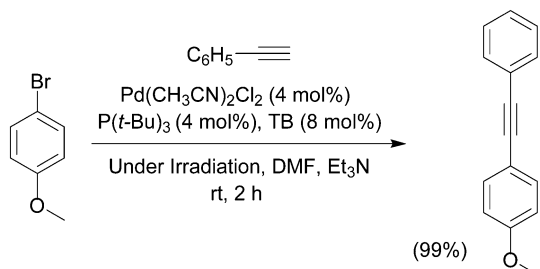
Scheme 12 Proposed intermediates in the photo generation and reaction of the vinylen benzenium ion.



Scheme 13 Photo-induced synthesis of electron-rich aromatic alkyne.

when the starting aromatic halides absorb poorly at 254/310 nm.

Further investigations in this area reported that activated aryl bromides efficiently couple with aryl or trimethylsilyl alkynes under visible light irradiation in the presence of a tris(bipyridine)ruthenium(II) complex (TB) at room temperature (Scheme 14).⁹⁰ The photo-excited TB complex serves as an electron donor to reduce the Pd(II) precursor, without forming

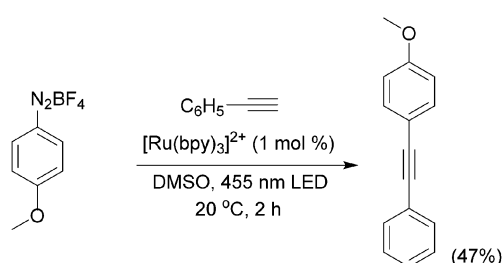


Scheme 14 Sonogashira reaction in the presence of tris(bipyridine) ruthenium(II) complex (TB), under irradiation of visible light.

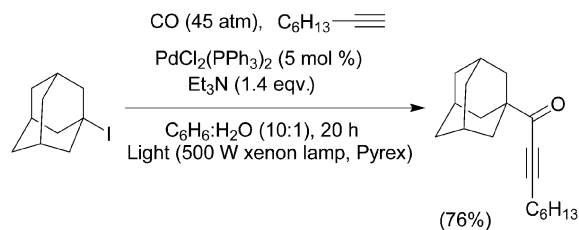
Pd(0)-phenylacetylide to promote the catalytic reaction. It should be noted that under the same conditions, the reaction of 4-chloroacetophenone and phenylacetylene afforded 4-acetyl-phenyl phenyl acetylene in 36% yield.

Another Sonogashira-type photocatalytic reaction involved the arylation of alkynes with diazonium salts by using 2,2'-bipyridine ruthenium complexes, [Ru(bpy)₃]²⁺ has been reported (Scheme 15).⁹¹ As observed from Scheme 15, the yield is moderated and in the original paper only two examples are reported, since the method was primarily developed for coupling diazonium salts with phenyl acetylene and its derivatives. The scope and possible application of this method is still to be developed.

Although the Sonogashira reaction is mainly associated with the coupling between a terminal alkyne and an aryl or vinyl halide, a three-component carbonylative Sonogashira reaction has recently been developed. In this procedure, an alkyl iodide, carbon monoxide (CO), and a range of alkynes are reacted in the presence of a catalytic amount of Pd in the presence of light, resulting in the formation of alkynyl ketones (Scheme 16).⁹² A



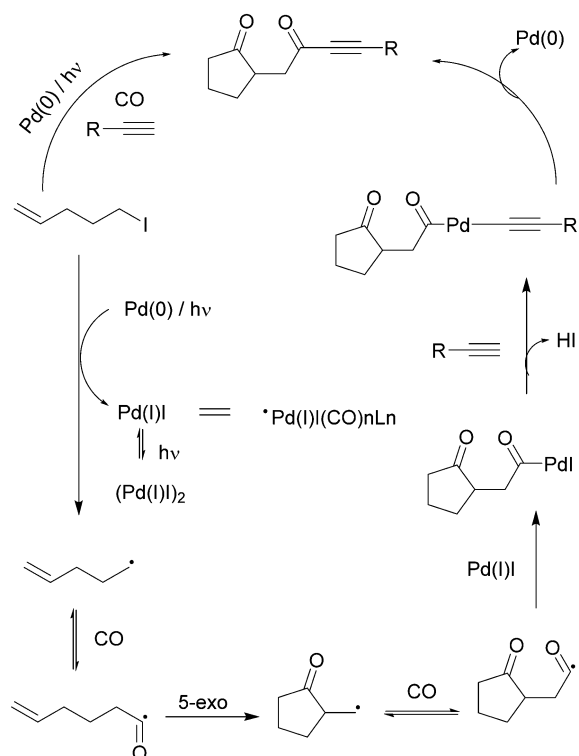
Scheme 15 Photocatalytic arylation of alkynes with diazonium salts.



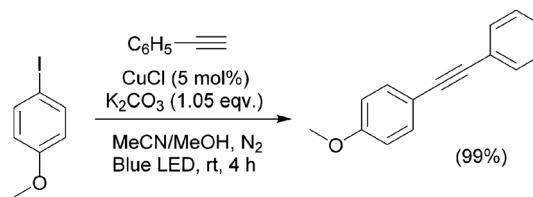
Scheme 16 Carbonylative Sonogashira reaction leading to alkynyl ketones.

variety of functional groups are tolerated in this reaction; however in this proposed methodology the reaction of aryl bromides and chlorides were not discussed. Mechanistic studies demonstrated that alkyl iodide reacts with Pd(0) under irradiation to afford alkyl radical and unfamiliar Pd(I)I⁹³⁻⁹⁵ via a one electron transfer process (Scheme 17). The resulting alkyl radical traps CO to form an acyl radical which then affords an alkyl radical through 5-*exo*-cyclization of the acyl radical. The acyl radical species generated from CO and the alkyl radical couples with Pd(I) iodide to lead to acylpalladium intermediate and gives the alkynyl ketone.

Recently, another study reported that simple Cu(I) chloride salt, in the absence of Pd and ligands, can catalyze the Sonogashira reaction under blue LED light irradiation at room temperature (Scheme 18).⁸⁴ In the process it is thought that Cu-acetylide is the key light absorbing species, and is the driving force for photo-induced Sonogashira reactions. The



Scheme 17 Possible mechanism for the carbonylative Sonogashira reaction.

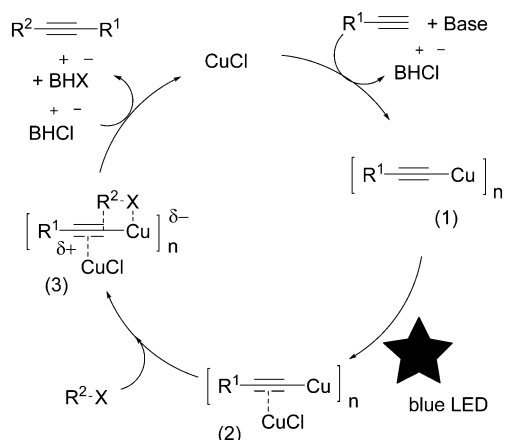


Scheme 18 Photo induced Sonogashira reaction of phenylacetylene with aryl iodides.

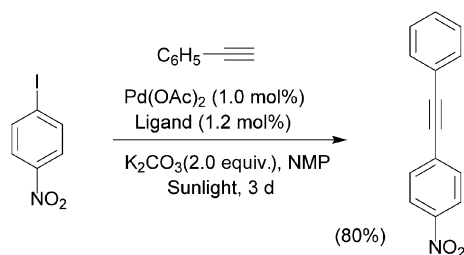
deprotonation by base and subsequent coordination by CuCl, results in *in situ* formation of Cu-phenylacetylide (1) (Scheme 19). As Cu-phenylacetylide can exist in a polymeric form,⁹⁶ the structure of (1) may be either a monomeric or a polymeric form or both. The direct photoexcitation of Cu-phenylacetylide ($\lambda_{\text{abs}} = 476 \text{ nm}$) leads to the formation of electron-deficient acetylene moiety *via* ligand to metal charge transfer (LMCT), which generates a partially positive charge at the acetylene site, and thus favors nucleophilic attack of electron rich substituted phenyl halides to form transition state intermediate (3) (Scheme 19), which leads to the direct formation of the C-C coupling products. It was observed that electron-rich aryl halides reacted significantly faster than the electron-deficient examples and aryl chlorides were unreactive in this process.

In contrast, another study demonstrated a simple way to utilize sunlight as a renewable energy source for Sonogashira reactions, without the need for any special apparatus or reaction conditions. The process involves using Pd-catalysts under irradiation with sunlight in the absence of Cu salt (Scheme 20).⁹⁷ Only three examples were reported, as the method was primarily developed for Mizoroki-Heck and Suzuki-Miyaura coupling reactions. Considering the high yields obtained for the three reported examples, this protocol should be more thoroughly investigated in order to uncover its full potential application and scope.

Although, UV/Vis light induced cross coupling works well for aryl bromides/iodides, in many cases it is ineffective for aryl



Scheme 19 Proposed mechanism for the visible light mediated CuCl-catalyzed Sonogashira reaction.



Scheme 20 Cu-free sunlight promoted Sonogashira reaction.

chlorides. Despite some limitations, these protocols have distinct advantages over the thermal alternatives, since the reactions are carried out at room temperature. Mild conditions such as low catalyst loading and high reaction yields at room temperature make these current approaches a very promising, industrially scalable green process as an alternative to the conventional high-temperature Sonogashira reactions, for the synthesis of natural products and medicinal compounds.

3.3. Nanoparticles as catalysts in Sonogashira reaction

Although, homogeneous transition metal catalysts remain the most active catalysts for Sonogashira reactions, efficient separation and recycling are still scientific challenges, because catalyst deactivation often occurs during the workup.⁹⁸ Additionally, homogeneous catalysts have a greater tendency to lead to metal contamination in the target products. To overcome these problems, metal nanoparticles (MNPs) have attracted much attention in the last several years. MNPs exhibit distinct physicochemical properties compared to their bulk counterparts due to their large surface-to-volume ratio, thus providing many highly active metal uncoordinated sites.⁹⁹ Furthermore, immobilizing MNPs onto solid supports can minimize atom/ion leaching from the particles.¹⁰⁰ Research into the preparation and use of supported MNPs as a catalyst in processes such as the Sonogashira reaction has been reported in recent years, which are particularly characterized by their catalytic activity, stability, recovery and reusability.

Activity: Generally, the performance of MNP catalysts for many different reactions is strongly dependent on the shape, size, dispersion and support and, consequently, on the synthesis methods, reducing agents, and stabilizers employed to make them. Transition MNPs are especially active catalytic systems due to their large surface area-to-volume ratios.^{101,102} It was observed that poly(vinylpyrrolidone) (PVP)-stabilized spherical-shaped palladium nanoparticles (Pd-NPs) with 2.1 nm in size were more active catalysts than PVP-stabilized tetrahedral-shaped platinum nanoparticles (Pt-NPs) with 5.0 nm in size.¹⁰³ As the nanoparticle size is associated with reactivity, small nanoparticles would allow more favorable oxidative addition of the metal to the carbon-halogen bond at the rim of the nanoparticle.¹⁰⁴

Stability: The most important aspect to discuss about the stability of metal NPs in the C-C cross-coupling reaction is associated with the 'Ostwald ripening' process,⁹⁸ which is the main reason behind cluster growth. This mechanism involves

the detachment of atoms from smaller clusters and the attachment of these atoms to the larger clusters that have lower surface energy. The high temperature reaction conditions can promote the increase of the nanoparticle size by Ostwald ripening process and lower the catalytic activity of the MNPs. Although the increase in size of the MNPs can be avoided by adding excess of stabilizers, such as trialkylammonium salts, ionic liquids, or poly(ethylene glycol) (PEG), which can act as ligands surrounding the dispersed nanoparticles and therefore minimizing their tendency to undergo agglomeration^{103–105} but the addition of excess stabilizers can make the catalyst less active. Furthermore, when excess of the reaction product is added into the reaction medium, the catalytic activity also decreases.

Recovery and Reusability: The recovery and reusability of the MNP catalysts are important issues in C-C cross-coupling reactions. The immobilization of colloidal MNPs on a solid support is a useful alternative as the catalyst can be recycled by simple filtration.¹⁰⁶ Recent findings revealed some nano-level and metalodendritic Pd-catalysts can be recovered by precipitation, ultrafiltration, or ultracentrifugation.^{107,108} As the reaction conditions for C-C cross-coupling usually involve high temperature for a long period of time, the growth of the MNPs by Ostwald ripening can be expected to continue during and after recycling of the catalysts. As a result, the catalytic activity of MNPs gradually decreases after each recycle.

3.3.1. Palladium nanoparticles as catalysts. Supported Pd nanocatalysts (Pd-NCs) (Fig. 5) are now one of the most accepted catalytic systems in "next generation" cross-coupling reactions and have many advantages over classical colloids. They are usually smaller (1–10 nm in diameter), with narrower size dispersion and their synthesis is reproducible with well defined composition and clean surfaces.¹⁰⁹ They are easily isolable and can be redissolved in aqueous and/or organic solvents. In terms of catalytic properties, Pd-NCs are usually more active and display reproducible activities and often high selectivity. Furthermore, Pd-NCs are also of great interest due to the potential for catalyst recovery and reuse.

A particular consideration concerning the mechanism of Pd-NPs catalyzed C-C coupling reaction when Pd-NPs are involved is whether the coupling is catalyzed by the nanoparticles themselves or by the molecular species. Thus, when Pd-NPs are used as catalysts, the leaching and the formation of catalytically active molecular species must be considered. According to this hypothesis, three possible mechanism have been proposed (Fig. 6): (a) Pd-NPs can act as a genuine heterogeneous catalyst, with the reaction occurring on the surface; (b) Pd-NPs can act as a reservoir of molecular species, and Pd(0) atoms can leach from the nanoparticles and enter into a homogeneous catalytic cycle; (c) the oxidative addition can occur at the nanoparticle surface followed by leaching of the formed [Pd(Ar)X] complexes, which then initiates another homogeneous catalytic cycle.

3.3.1.1. Palladium nanoparticles with copper co-catalysts. Although one of the main interests in using Pd-NPs in the Sonogashira process lies in their high reactivity, and therefore the suitability of performing Cu and ligand-free couplings, there are examples where Cu(I) and a phosphine ligand have

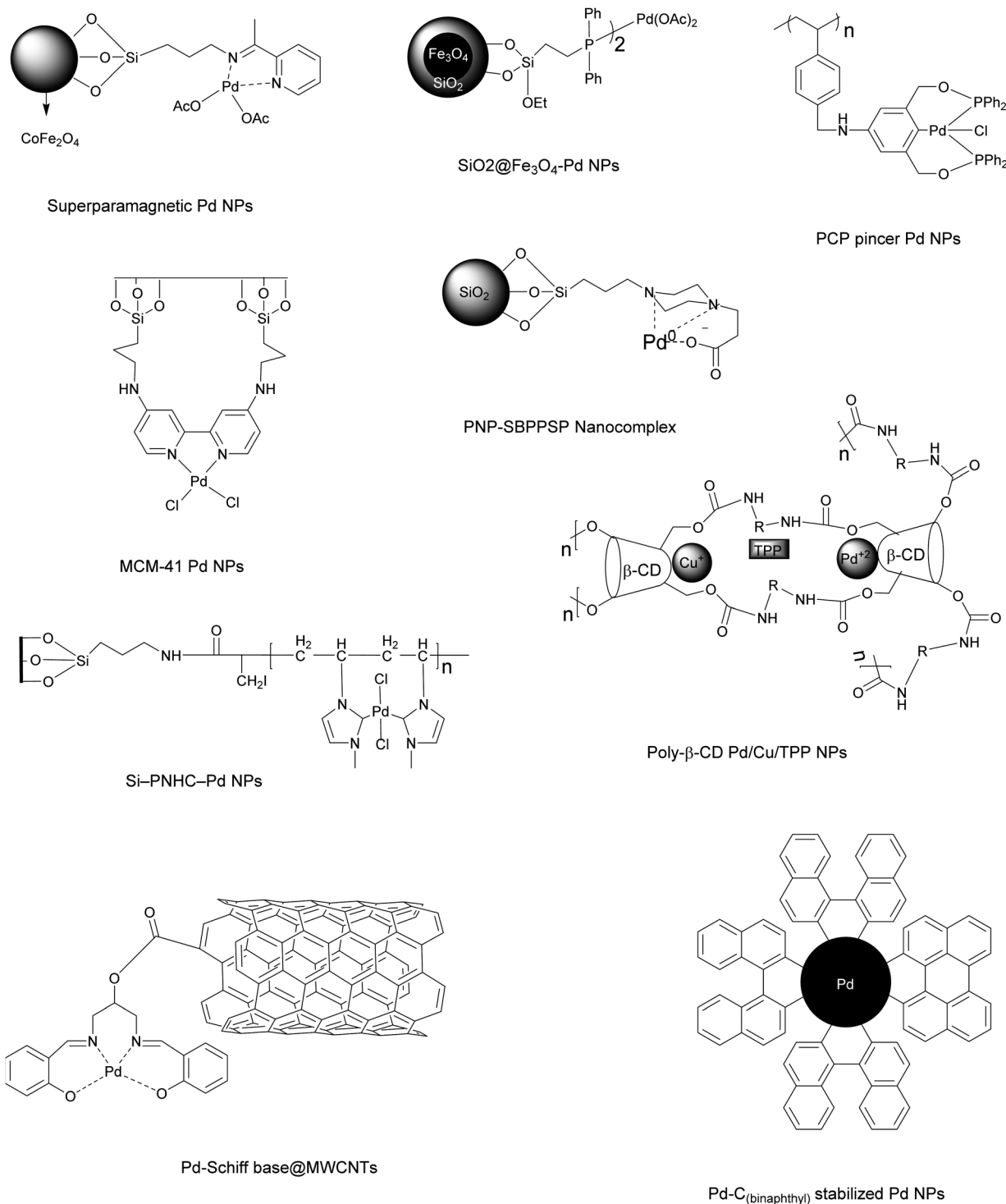


Fig. 5 Examples of recently developed supported palladium nanoparticles (Pd-NPs).

been added to a homogeneous reaction catalyzed by Pd-NPs. The Sonogashira reaction was efficiently catalyzed with receptor-modified β-cyclodextrin-capped Pd-NPs (β-CD/Pd)¹¹⁰ and co-catalyzed with CuI in aqueous solution at room temperature without phosphine ligand through host-guest recognized catalytic process (Scheme 21). The Sonogashira reaction of aryl bromides was relatively slow with lower yields

than those reported for the corresponding aryl iodides under the same conditions; however, eliminating the organic solvent in the Sonogashira reaction represents a significant advancement in the development of a greener process.

Recently it has been demonstrated that nano-sized MCM-41-Pd catalyst¹¹¹ effectively catalyze the Sonogashira reaction of various aryl and heteroaryl halides with phenyl acetylene

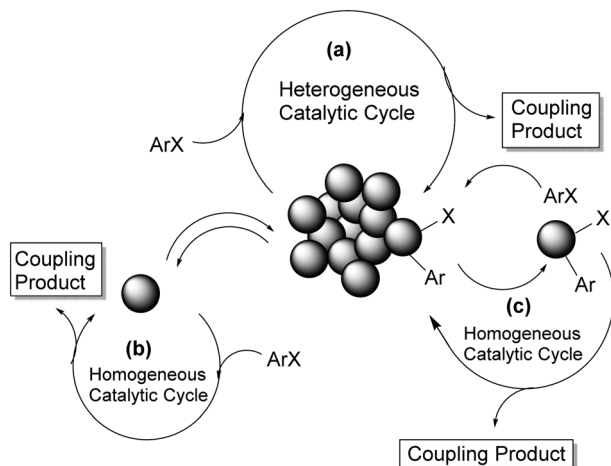
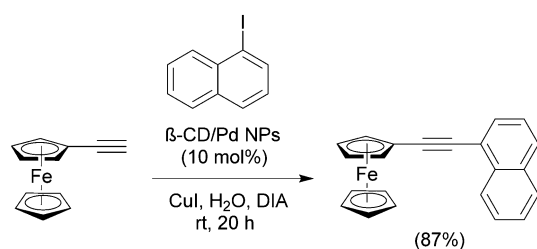
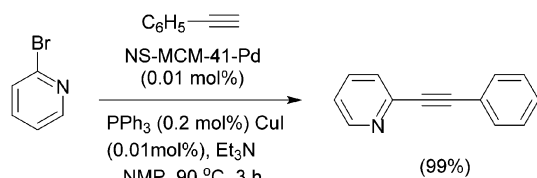


Fig. 6 The three possible mechanisms for the Pd-NPs catalyzed C-C coupling reaction.¹⁰⁰



Scheme 21 The aqueous Sonogashira reaction catalyzed by β -CD/Pd-NPs.

(Scheme 22) and alkynols. Reactions of aryl iodides with several EWGs and EDGs proceeded well with the use of 0.01–0.1 mol% of NS-MCM-41-Pd, and the Sonogashira reaction of aryl bromides under the same conditions did not afford any product. Replacing the TEA solvent by NMP resulted in product formation with a yield of 30% at an elevated temperature. A yield of up to 56% could be achieved by performing the reaction in toluene at 100 °C for 24 h. Aryl chlorides are completely inert under the aforementioned conditions. However, the major advantage of this catalyst is that the short and highly-connective wormhole-like channels of nanosized MCM-41 lead to facile exchange of reactants, salts and products throughout the nanochannels, avoiding saturation of activity. In addition, the nanosized low Pd loadings (0.01 mol%) MCM-41-Pd catalyst was recovered by centrifugation of the reaction solution and re-used in further runs without significant loss of reactivity.



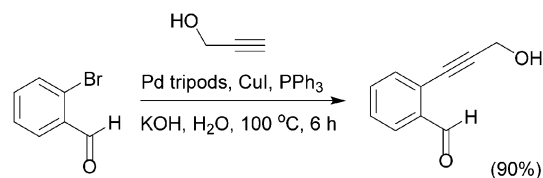
Scheme 22 Nanosized MCM-41 supported Pd catalyzed Sonogashira reaction.

Further studies have shown that palladium–polypyrrole (Pd–PPy) nanoparticles¹¹² in combination with a Cu(I) salt, can be efficiently used as a catalyst for the Sonogashira reaction of iodobenzene with phenylacetylene, in the presence of sodium carbonate with NMP as the solvent. In this procedure the addition of DMF or DMSO facilitates the use of lipophilic organic substrates and inorganic salts used as bases. This typical Pd–NC is an active catalyst for the Sonogashira coupling of iodo- and bromoarenes but not for the chloroarenes. Another study demonstrated that recyclable and highly active Pd tripods nanoparticles¹¹³ catalyzed the Sonogashira reaction in water as a sole solvent (Scheme 23). Their use as recyclable catalysts was investigated in the reaction between iodobenzene and phenyl acetylene to form diphenyl acetylene. The product yield was 93% for the first run after 6 h of reaction at 100 °C, and 91% and 90% for the second and third cycles, respectively.

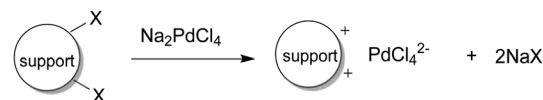
The reaction carried out with a variety of aryl iodides and bromides containing EDGs and EWGs in the presence of CuI as a co-catalyst, triphenyl phosphine as the ligand and potassium hydroxide as the base. In general, the Pd tripods were successfully catalyzed the Sonogashira reaction with excellent product yields ranging for 85–93%. The versatility of the Pd tripod nanocrystals for a broad scope of Sonogashira reactions and their easy separation from the products implies that this protocol have great potential to be explored in synthetic work and they may be employed for producing many important bioactive molecules.

3.3.1.2. Palladium nanoparticles in copper-free reaction conditions. Cu-free and ligand-free Sonogashira reactions have been carried out using different procedures in order to stabilize the Pd-NPs. In 2002, the first example of Cu and ligand-free heterogeneous immobilized layered double hydroxide and Merrifield resin supported nanopalladium [LDH-Pd(0)]¹⁰⁶ catalysts exhibited higher activity and selectivity in the Sonogashira reaction of electron-poor and electron-rich chloroarenes over the homogeneous PdCl₂ system. The catalyst is quantitatively recovered from the reaction by a simple filtration and reused for a number of cycles with almost consistent activity in the coupling reactions (Scheme 24).

Stable and crystalline clusters of Pd(0)-NPs¹¹⁴ can also catalyze the Sonogashira reaction at room temperature in the



Scheme 23 Pd Tripods catalyzed Sonogashira reaction.



Scheme 24 Synthesis of exchanger-PdCl₄ nano catalyst.

presence of the ionic liquid 1,3-dibutylimidazolium tetrafluoroborate [(bbim)BF₄] (Scheme 25). The reaction is carried out under ultrasound irradiation with enhanced reaction rates and excellent chemoselectivity. In this procedure neither Cu co-catalyst nor phosphine ligands are required. It was observed that the process tolerates both EDGs and EWGs in the aryl halides as well as terminal alkynes. However, in the case of less reactive bromobenzenes, the homocoupled product was observed to an extent of 6–7% and aryl chlorides are inert in this catalytic system.

Notably, in the absence of ligand, Cu and amine, a recyclable poly(vinylpyrrolidone) (PVP) supported nanosized Pd(0) metal colloids catalyzed Sonogashira reaction of aryl iodides and bromides with terminal alkynes was developed without significant loss of the catalytic activity for eight consecutive cycles (Scheme 26) although, decreases in reaction yields were observed after eight cycles. The electron-neutral, electron-rich, and electron-poor aryl iodides reacted with terminal alkynes smoothly. Regardless of their electronic character, both of the aromatic terminal alkynes and aliphatic terminal alkynes components coupled with aryl iodides. For electron-rich aryl bromides, relatively low yield was obtained under the reaction conditions.¹¹⁵

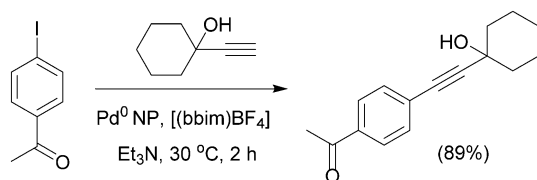
Recently, PVC (polyvinyl chloride) supported Pd-NPs¹¹⁶ were also explored for the Sonogashira reaction in water as a solvent. In this case the catalyst only works for the aryl iodides substituted with several EWGs and EDGs. The catalyst can be used in five successive runs with no loss in its reactivity in Sonogashira reactions which confirm the stability of Pd-PVC composite. The slight decrease observed in the yield during the recycling experiment can be accounted to agglomeration of metal nanoparticles within the PVC matrix during the course of the reaction. In the same year, another recent study revealed, PVC supported Pd-NPs [PVC-dtz-Pd(II)]¹¹⁷ (Scheme 27) are highly functionalized for Sonogashira reaction of versatile aryl

halides including aryl chlorides with different EDGs and EWGs in aerobic conditions. However, in the case of aryl chlorides the product yield is low to moderate, 43–64%. The solid catalyst was recovered by separation of the organic compounds from the reaction mixture by extraction, and the recovered catalyst was repeated for five cycles; some decrease in the catalytic activity was observed with metal leaching.

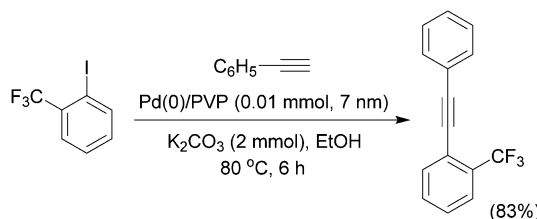
High leaching-resistant Pd-NPs on carbonized electrospun polyacrylonitrile (PAN) nanofibers (Pd-NPs/CENFs)¹¹⁸ showed good catalytic activity in the Sonogashira reaction of iodo-benzene and phenylacetylene in liquid-phase. The catalyst was retrieved after each run by filtration, washing and drying under vacuum and repeatedly used for ten times without significant loss of reactivity. It was also reported that Pd nanoclusters,¹¹⁹ synthesized by a simple method in a 1 : 1 mixed solvent system of MeCN–MeOH without any stabilizer were found to catalyze the Cu, amine and ligand free Sonogashira reactions in open air; however this catalytic system only works for aryl iodides (Scheme 28). In this case, the material existed as leached Pd nanoclusters during the reactions but it was unclear if Pd was redeposited at the end of the reaction.

Another study illustrated that PEG stabilized Pd-NPs,¹²⁰ synthesized from the reaction of a Fischer carbene complex of tungsten with K₂PdCl₄ at room temperature, displayed high catalytic activities in the Sonogashira reaction. Under the conditions described, water is the sole solvent and no Cu(I) salt, phosphine ligand or amine are required. Although the reaction proceeded well with aryl iodides, no coupling product was observed with activated bromides even at elevated temperatures under a wide range of conditions. It was also reported the application of Pd-NPs supported on MOF-5¹²¹ in the Cu and amine-free Sonogashira reaction in the presence inorganic base in methanol. Recently, the immobilized, homogeneous and super paramagnetic Pd-NPs¹²² (Scheme 29) was used as an efficient catalyst for the Cu-free Sonogashira reaction without added phosphine ligands. The catalyst was recovered by simple magnetic decantation and reused several times without significant degradation in catalytic activity. The presence of EWGs on the benzene ring significantly accelerated the Sonogashira reaction. Although, this catalytic system successful for activated aryl bromides, chlorobenzene was inactive in the reaction with terminal alkynes.

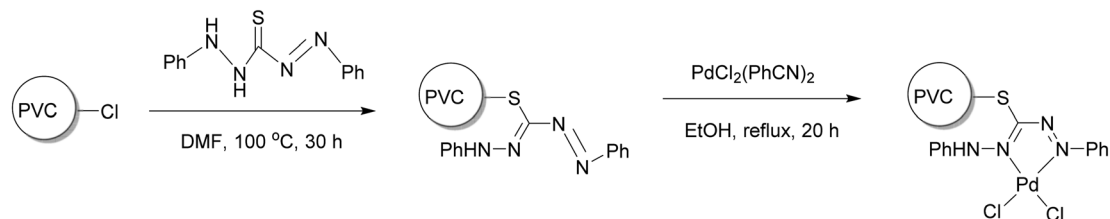
It has been further demonstrated that organosilica xerogels doped with nanostructured Pd(0) can be used as stable heterogeneous and reusable catalysts,¹²³ suitable for performing Sonogashira reactions in high selectivity without the need to exclude air or moisture; however the catalyst was successful only for aryl iodide. Reusing the catalyst in seven consecutive cycles did not result in any loss in catalytic activity with negligible (less than 0.2 ppm) leaching of Pd and Si. Complete substrate conversion was obtained even after the seventh cycle, with 99.5% selectivity. Notably, another study developed an air and moisture-stable and easily recoverable silica-coated magnetic nanoparticle-supported Pd catalyst (SiO₂@Fe₃O₄–Pd)¹²⁴ (Scheme 30) for the Sonogashira reaction in Cu-free condition (Scheme 31) but the catalytic system was no longer effective for the couplings of aryl chlorides. More recently,



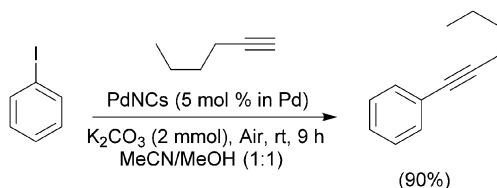
Scheme 25 Sonogashira reaction catalyzed by Pd(0) nanoparticles under ultrasound irradiation.



Scheme 26 PVP supported nanosized Pd catalyzed Sonogashira reaction.

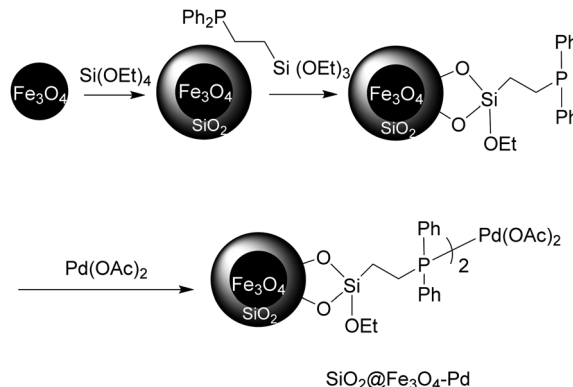


Scheme 27 Synthesis of PVC-dtz-Pd(II) catalyst.

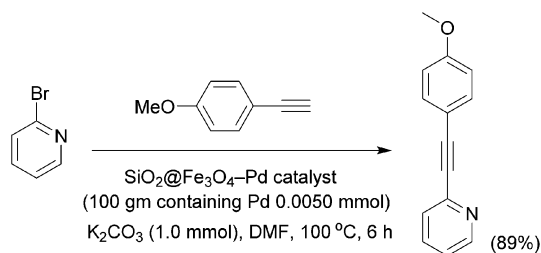
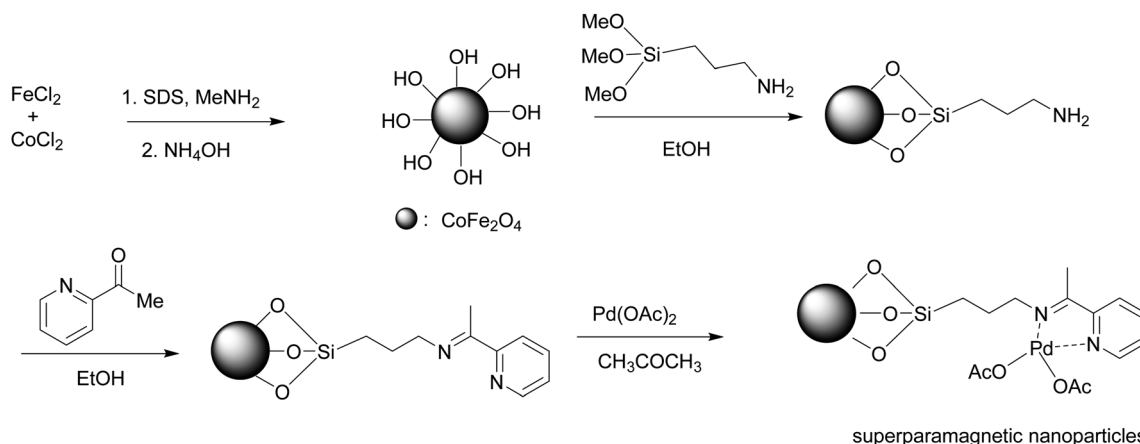


Scheme 28 Palladium nanocluster-catalyzed Sonogashira reaction.

superparamagnetic nanoparticles functionalized with Schiff base immobilized Pd(II) complex $[\text{Fe}_3\text{O}_4@\text{SiO}_2/\text{Schiff base-Pd(II)}]^{125}$ were used as the catalyst for Cu and phosphine ligand-free Sonogashira reaction. Notably, this new catalyst also works for the coupling of aryl chlorides with excellent yield (79–92%). In another study, $[\text{Fe}_3\text{O}_4@\text{SiO}_2\text{-polymer-imid-Pd}]$ nanocatalyst¹²⁶ has been efficiently applied for the Sonogashira reactions. The coupling reaction of phenylacetylene with both electron-releasing and electron-withdrawing aryl iodides and bromides afforded the coupling products in good yields, although aryl chlorides are less reactive. In contrast, a recent research developed a super paramagnetic nanocatalyst for Sonogashira reaction where palladium supported on nano-magnetite ($\text{Pd}@\text{Fe}_3\text{O}_4$) without SiO_2 support.¹²⁷ The catalyst was separated by a simple external magnetic field and the recycled catalyst was reused six times with no loss of activity. The electron-neutral, electron-rich and electron-poor aryl halides reacted with phenylacetylene smoothly to generate the



Scheme 30 Preparation of the magnetic nanoparticle-supported palladium catalyst.

Scheme 31 $\text{SiO}_2@\text{Fe}_3\text{O}_4\text{-Pd-NPs}$ supported Sonogashira reaction.

Scheme 29 Synthesis of the palladium catalyst immobilized on super paramagnetic nanoparticles.

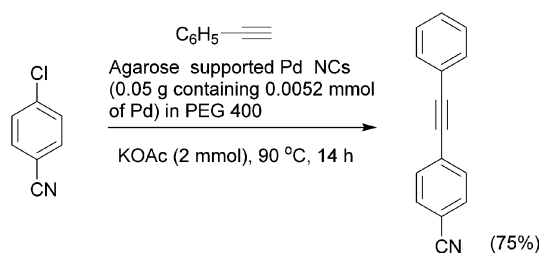
corresponding cross-coupling products, although, aryl chlorides were not successful in this protocol.

Recently, agarose-supported reusable Pd-NPs¹²⁸ have been developed for the Sonogashira reaction. In this protocol, PEG-400 is used as a green solvent under Cu and phosphine-free conditions, which caused the regular dispersion of the Pd-NPs on the surface of the agarose (Scheme 32). Herein, a range of aryl iodides, bromides and chlorides, bearing EWGs and EDGs, reacted with phenyl acetylene successfully. The recovered catalyst was recycled for five successive runs with a rather sharp decrease in the catalytic activity from the fourth run. The leaching of the catalyst into the reaction mixture was also determined by ICP analysis for the first run of recycling to be less than 2% after cooling the reaction mixture to room temperature.

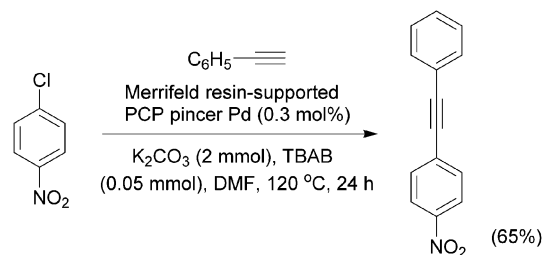
The Pd nanoparticle catalysts supported on partially reduced graphene nanosheets (Pd/PRGO),¹²⁹ synthesized by pulsed laser irradiation of aqueous solutions of graphene oxide and palladium ions, was successfully utilized in the Sonogashira reaction under ligand free microwave irradiation conditions. With more challenging substrate such as electron poor aryl chlorides, conversions of 10–15% were reported. Another recoverable and reusable high TON catalytic system of bisphosphinite PCP pincer Pd complex (Scheme 33) based on Merrifield resin¹³⁰ was

developed. This method exhibited good catalytic activity in Sonogashira reactions with different EWGs and EDGs connected aryl halides including inactive aryl chlorides under Cu and amine-free condition (Scheme 34). The recovery and reusability of the supported catalyst was investigated using iodo-benzene and butyl acrylate as model substrates. The recovered catalyst was reused in ten subsequent reaction cycles and the catalyst retained its catalytic activity. A recent report demonstrated that the amido pincer complex is a competent catalyst precursor for the Sonogashira reactions of aryl halides with terminal alkynes under mild conditions.¹³¹ This catalysis is compatible with a variety of functional groups including alkyl, alkenyl, (hetero)aryl, silyl, fluoro, alkoxy, ketone, nitro, *etc.*

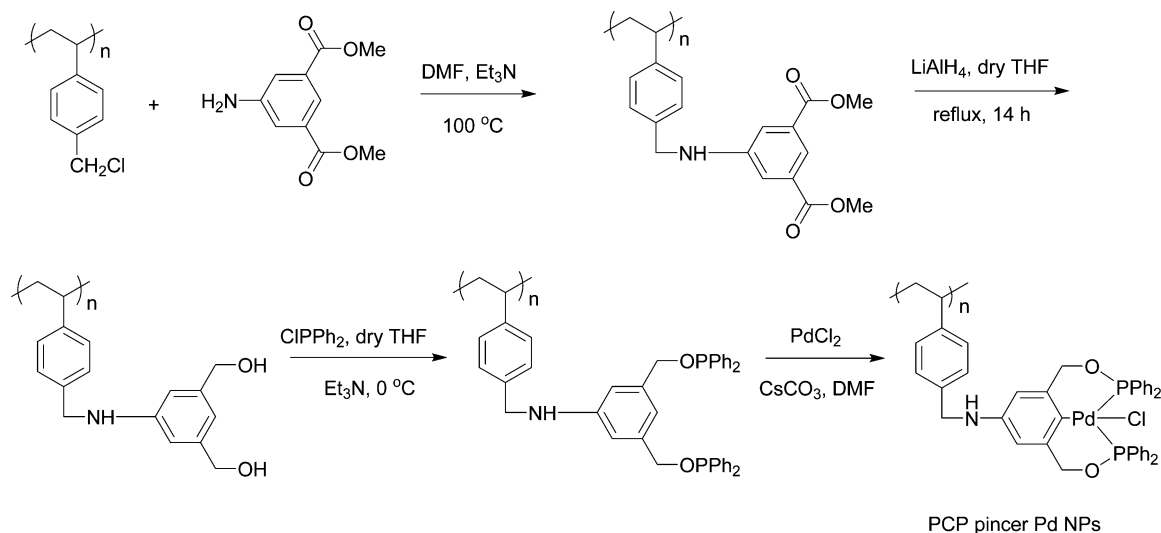
Another heterogeneous, recyclable, high TON and TOF catalytic system based on Pd-NPs supported on polymeric N-heterocyclic carbene grafted silica (Si-PNHC-Pd) has also been developed.¹³² This catalyst was reused in twelve subsequent reactions and the catalyst retained its catalytic activity. Under such conditions excellent results in the Sonogashira reaction with various aryl halides including some aryl chlorides in the presence of a catalytic amount of TBAB were achieved. A heterogeneous stabilized retrievable Pd catalyst system has been developed based on immobilization of Pd-NPs on silica-bonded *N*-propylpiperazine sodium *N*-propionate (PNP-



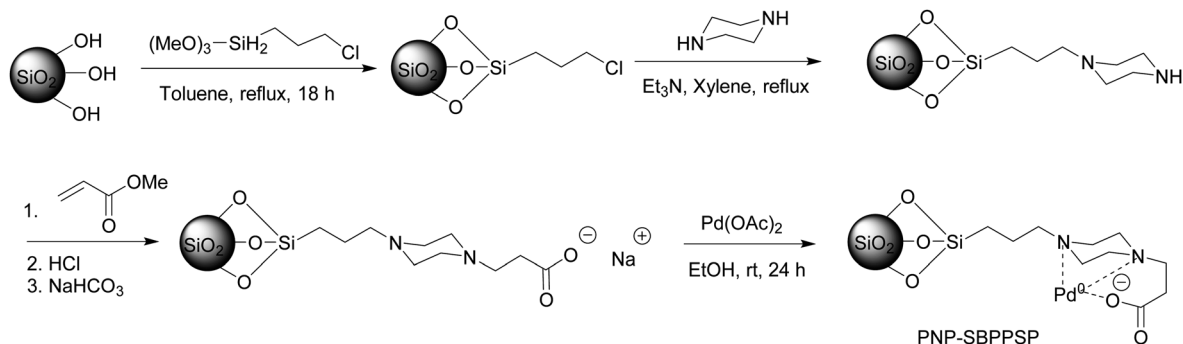
Scheme 32 Agarose supported Pd nano catalyzed Sonogashira reaction in PEG-400.



Scheme 34 Merrifield resin-supported PCP pincer Pd catalyzed Sonogashira reaction.



Scheme 33 Synthetic strategy for the preparation of modified Merrifield resin-supported PCP pincer palladium nanoparticles.



Scheme 35 The synthetic route for preparation of PNP-SBPPSP.

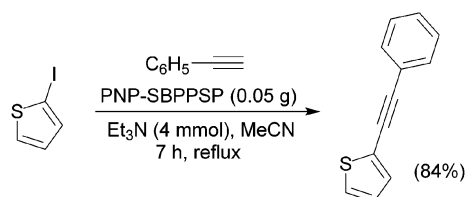
SBPPSP) substrate¹³³ (Scheme 35). The catalyst can be recovered by simple filtration. The recycled catalyst was reused four times without loss of activity. This system efficiently catalyzes the Sonogashira reactions in Cu-free conditions including a range of aryl iodides, bromides and chlorides, bearing EWGs and EDGs (Scheme 36).

Recently a new method involving *in situ* generation of recyclable Pd-NPs into the nanopores of modified montmorillonite¹³⁴ was developed. The catalytic activity of such nanoparticles in the Sonogashira reaction was high and 100% selectivity was observed when MeCN was the solvent in the presence of triethylamine and under Cu-free conditions. In another study, the use heterogeneous Pd/graphene (Pd/G) nanocomposites¹³⁵ was reported. In this method, the catalyst showed good efficiency in Sonogashira reactions and the ability to recover and recycle the catalyst was an advantage. The activity of the catalyst was attributed to surface defects on graphene that act as nucleation and growth sites for Pd-NPs and facilitate higher dispersions of Pd-NPs. Notably, in another study, a novel Pd-containing 2D-hexagonal periodic mesoporous organosilica material (Pd-LHMS-3)¹³⁶ showed good catalytic activity in Sonogashira reaction in the presence of water as the sole solvent and hexamine as base without Cu co-catalyst (Scheme 37). The

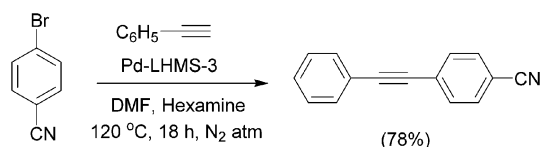
main advantages of this catalyst is the presence of many donor sites, which serve as better chelators for the Pd-sites and thereby minimizes its possible leaching into solution. Upon completion of the reaction, the catalyst can be recovered by simple filtration and reused up to four times without any significant loss of reactivity, although conversion levels decreased significantly after the fourth catalytic cycle. EWGs and EDGs containing aryl iodides and sterically hindered *ortho*-substituted aryl iodides and heteroaryl iodides are well tolerated; although the Sonogashira reaction of aryl bromides did not occur in water and in the presence of hexamine, aryl bromide react with phenyl acetylene in DMF under nitrogen atmosphere.

Interestingly, the air-moisture stable and recyclable Pd(π)-Schiff base complex anchored to multi-walled carbon nanotubes (Pd-Schiff base@MWCNTs)¹³⁷ behaves as a very efficient heterogeneous catalyst in the Sonogashira reaction in aqueous media. The coupling reaction of bromobenzene with phenylacetylene were very slow under the same reaction conditions giving trace amount of cross-coupled product after 24 h of reaction time and the coupling reaction of chlorobenzene with phenylacetylene did not occur at all. A new class of Pd-NPs,¹³⁸ stabilized by Pd-C(binaphthyl) (Scheme 38) covalent bonds which helped the catalyst to retain its activity and particle size and restricts agglomeration, was efficiently applied as reusable catalysts for Sonogashira reactions with high turnover. After completion of the reaction, the catalyst was successfully recovered and reused for the next 2–3 cycles. EWGs and EDGs containing aryl iodides are well tolerated and provided excellent yields; although the Sonogashira reaction of aryl bromides and chlorides did not occur.

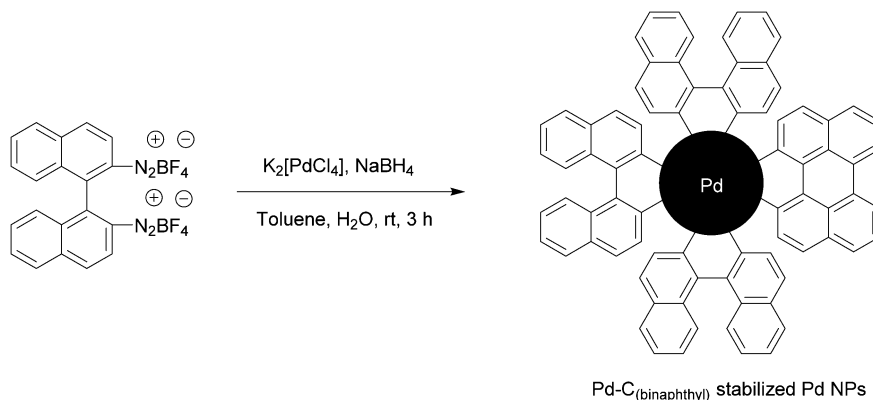
The application of Pd-NPs (2–6 nm) supported on pectin¹³⁹ exhibited high catalytic efficiency and air stability in the Sonogashira reaction without Cu, amine and phosphine was also reported (Scheme 39). Although in the case of aryl iodides it was found that the activity of the catalyst was completely retained after two recycled runs, a gradual decrease in catalytic activity was observed in some cases. In the case of less reactive electron-rich aryl bromides such as 4-bromotoluene and 4-bromoanisole, the coupling reaction carried out over a longer time period. Very strong EWG (such as NO₂) on the aromatic ring led to a high yield after shorter reaction time. Heterocyclic aryl bromides are also suitable substrates for this reaction. The



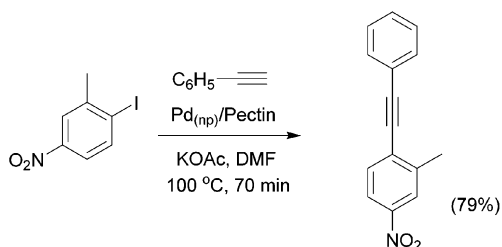
Scheme 36 PNP-SBPPSP catalyzed Cu-free Sonogashira reaction.



Scheme 37 Cu-free Sonogashira cross-coupling reaction over the Pd-LHMS-3 catalyst.



Scheme 38 Synthesis of Pd–NPs stabilized by Pd–C_(binaphthyl) covalent bonds.



Scheme 39 Pd_{np}/pectin catalyzed Sonogashira reaction.

coupling of aryl chlorides could be performing at 100 °C with good yields.

Very recently, a new polystyrene supported thiopseudourea–Pd(II) complex (Fig. 7) was found to be an active catalyst for the Cu-free Sonogashira reaction of different aryl halides.¹⁴⁰ The catalyst was easily recoverable from the reaction mixture by simple filtration and reusable up to ten times without significant loss of activity. However, the catalyst only works for EDGs and EWGs containing aryl iodides.

The Sonogashira coupling reaction between a range of aryl halides and phenylacetylene were carried out in the presence of a new Pd supported on ZnO nanoparticles (nano Pd/ZnO).¹⁴¹ It has been found that aryl iodides with a variety of EWGs and EDGs reacts well affording coupling products. On the other hand the reactivity of aryl bromides and chlorides with EWGs

was higher than those bearing EDGs. Catalytic recycling experiments were carried out using a Sonogashira reaction of 1-(4-bromophenyl)ethanone with phenyl acetylene as a model reaction. The catalyst could be recovered from the reaction mixture by diluting with EtOAc and centrifugation to separate the catalyst. Only 0.01% of the Pd metal after is leached out from the catalyst surface after five cycles. Recently, a tris-imidazolium iodide stabilized Pd–NPs catalyzed the Sonogashira reaction with activated and deactivated iodoarenes has been described.¹⁴² The reaction is also successful for activated aryl bromides, whereas nonactivated aryl bromides don't react. The catalyst proved to be tolerant to a range of functional groups, with the expected products being obtained in moderate-to-good yields. A convenient method for the synthesis of PdNPs–PPS [PPS = poly(1,4-phenylene sulfide)]¹⁴³ embedded into a polymer matrix *via* the thermolysis of palladium acetate is described which can be utilized as an efficient heterogeneous nanocatalyst for the acyl Sonogashira reaction under Cu-free reaction conditions (Scheme 40). The catalyst was recycled up to four times consecutively without any substantial change. The acyl Sonogashira reaction was examined through a variety of aryl/aliphatic acid chlorides with different aromatic/aliphatic terminal alkynes and was obtained in good yield.

Significantly, another novel Pd–NPs supported on two Smopex® metal scavengers (Pd/Smopex®-234 & Pd/Smopex®-111) can be successfully used for the three component carbonylative Sonogashira reaction affording the corresponding α,β -alkynyl ketones with high chemoselectivity.¹⁴⁴ It has been reported that Pd/Smopex®-234 is a good catalyst for the reactions of phenylacetylene with aryl iodides bearing both EDGs (Me, OMe, Ph) and EWGs (Cl) in *ortho* or *para* positions. All reactions yielded the carbonylated coupled products with high

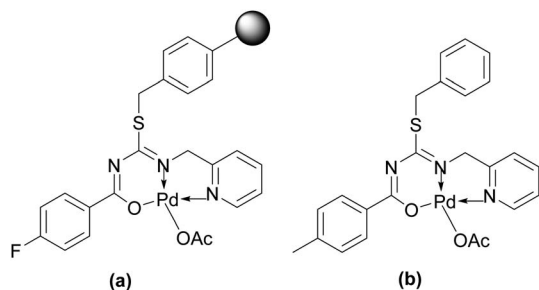
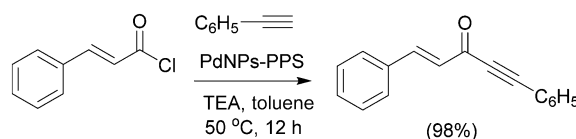


Fig. 7 Polymer supported (a)/non-supported (b) thiopseudourea–Pd(II) complexes.



Scheme 40 Sonogashira reaction using acylchlorides as substrate catalyzed by PdNPs–PPS.

chemoselectivity (82–93%). However, no conversion towards the alkynyl ketones was observed in the case of aryl bromides. The results obtained in the hot filtration test, the very low value (0.8%) of the Pd leached into solution during the reaction and the possibility of recycling the catalyst with no loss of activity seem to indicate heterogeneous behavior of Pd/Smopex®-234. Very recently, a new class of Pd-NPs grafted at the surface of Co-containing metal–organic framework material MCoS-1 (Pd(0)/MCoS-1)¹⁴⁵ has been developed which revealed good catalytic activity for Sonogashira reactions in water without Cu co-catalyst (Scheme 41). The catalyst is easily recoverable by simple filtration and can be reused for six times without appreciable loss of catalytic activity. Importantly, results of the hot filtration test suggested that Pd did not leach out from the solid catalyst during the reaction. The aryl halides with EWGs gave excellent yields, while aryl halides with EDGs gave the coupling products in slightly lower yield. The *ortho*-substituted aryl iodides gave lower yield than the *para*-substituted ones, presumably due to steric effects. The coupling reaction of aryl bromides required longer reaction times and chlorobenzene gave no desired product.

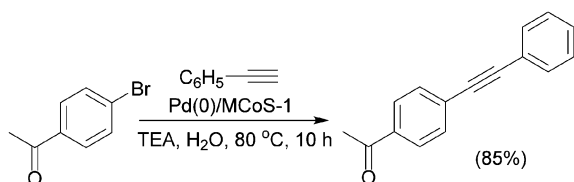
3.3.1.3. Palladium containing bi-/trimetallic nanoparticles. The search for more efficient catalytic systems that combine the advantages of both homogeneous and heterogeneous catalysts is one of the most exciting challenges in catalysis research. In this regard, the Ni/Pd dual core nanoparticle¹⁴⁶ catalyst, synthesized from the consecutive thermal decomposition of metal–surfactant complexes was applied to various Sonogashira reactions including sila-Sonogashira protocol (Scheme 42) and showed significantly enhanced catalytic activity when compared to normal Pd-NPs. Although, this semi-heterogeneous nanoparticle catalyst was recycled and reused five times without loss of catalytic activity for active bromoaryl substrates, it was inactive for chloroaryl substrates. In another aspect, a Pd/Cu co-catalyzed sila-Sonogashira reaction of (hetero)arylethynylsilanes with a variety of activated and deactivated (hetero)aryl

bromides in the presence of a phase-transfer catalyst (benzyl(tri-*n*-butyl)ammonium chloride) has been developed.¹⁴⁷

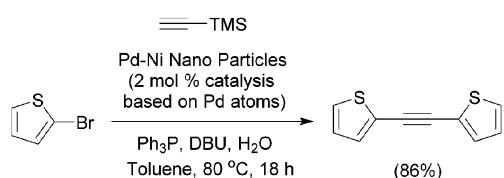
Pd containing Au–Ag–Pd triple core trimetallic nanoparticles¹⁴⁸ were synthesized in the presence of cetyltrimethylammonium bromide as the capping agent. The material exhibited better catalyst activity in Sonogashira reactions compared to Pd-NPs under aerobic conditions. This activity is possibly due to the concerted electronic effects of the Au–Ag core onto the Pd shell atoms. The catalyst can be recovered by precipitation and ultracentrifugation and the recovered catalyst reused for the next reaction cycle. Notably, another study reported a sustainable protocol to synthesize the poly- β CD Pd/Cu/TPP nanoparticles¹⁴⁹ (Scheme 43) which has moderate catalytic activity in Sonogashira reactions with very low metal leaching. The use of water or glycerol as a solvent rendered product purification easier and also allowed the catalyst to be recycled *via* simple filtration and washing. Another Cu-supported bimetallic nanopalladium catalyst (nano-Pd/PdO/Cu) was shown to be a selective catalyst for the Sonogashira reaction of different aryl halides.¹⁵⁰ However, electron-deficient aryl halides favor this reaction, while electron-rich aryl halides are less prone to the process. Up to two cycles of full catalyst performance (100% conversion) can be attained, and a very high deactivation in a third cycle (5% conversion) was reported. Very recently, another Pd/Cu bimetallic nanoparticle embedded in macroporous ion-exchange resins has been developed and successfully applied for the Sonogashira reaction although it exhibited poor catalytic activity with deactivated aryl bromides.¹⁵¹ The recovered catalyst by simple filtration was reused up to four times without losing its activity.

In a recent study, an efficient heterogeneous catalyst based on Pd–Co alloy nanoparticles¹⁵² supported on polypropylenimine dendrimers was developed. The nanoparticles are grown on graphene nanosheets, and can be efficiently used for Sonogashira reactions under Cu and solvent-free conditions using ultrasound irradiation at room temperature. The catalyst can be easily recovered by simple filtration and reused six times without significant loss of activity (Scheme 44). From the reported results, the Sonogashira reaction of terminal alkyne with aryl halides including aryl chlorides possessing both EDGs and EWGs proceeded in high yields. It has been very recently shown that graphene (G) supported Pd–Co bimetallic NPs (Pd–Co/G) can successfully catalyze the Sonogashira reaction.¹⁵³ Although, Pd–Co/G exhibited high selectivity, the catalyst is limited to aryl iodide and bromide. The catalyst can be easily recovered by a magnet due to their magnetic nature and reused five times without loss of significant activity.

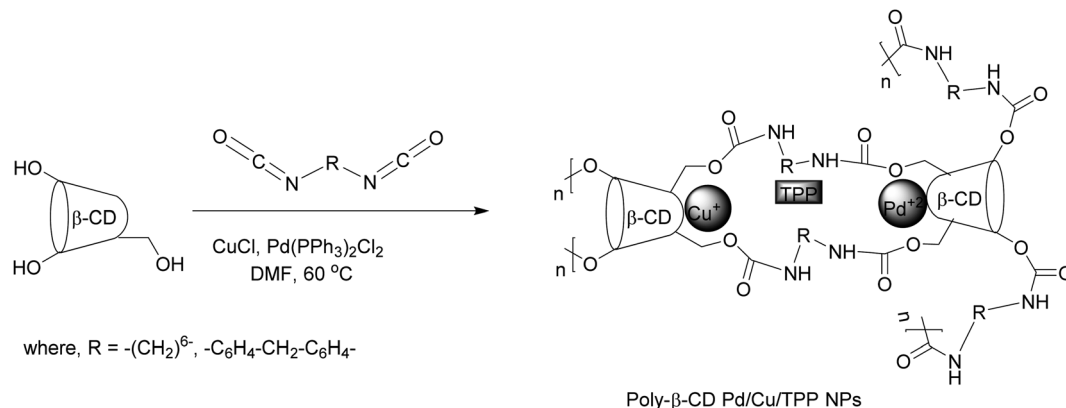
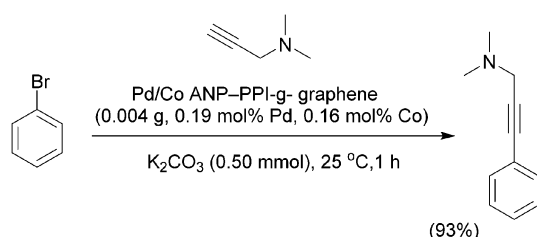
A magnetically separable and recyclable bimetallic heterogeneous Pd/Fe₃O₄ catalyst¹⁵⁴ showed highly efficient catalytic activity for the carbonylative Sonogashira reaction under phosphine-free conditions. During the reaction between aryl iodides and terminal alkynes, the Pd is leached and served as the active catalyst, and at the end of the reaction, Pd is redeposited onto the support. Recently, a convenient bimetallic Pd/Fe₃O₄ heterodimeric nanocrystal¹⁵⁵ catalyst system for Sonogashira reaction was developed (Scheme 45). In this process the



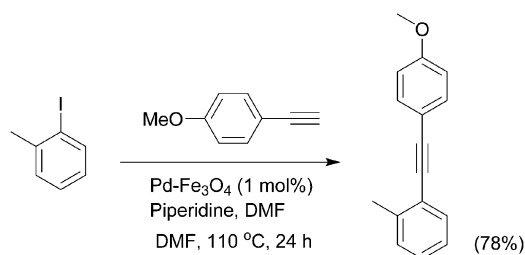
Scheme 41 Sonogashira reaction in the presence of Pd(0)/MCoS-1 catalyst.



Scheme 42 Sila-Sonogashira reaction catalyzed by Ni/Pd core/shell nanoparticles.

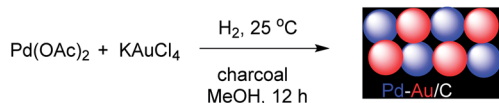
Scheme 43 Preparation of poly- β CD/Pd/Cu nanoparticles.

Scheme 44 Pd-Co ANP-PPI-g-graphene hybrid material catalyzed Sonogashira reaction.

Scheme 45 Pd-Fe₃O₄ heterodimeric nanocrystal catalyzed Sonogashira reaction.

catalyst could be easily separated by an external magnet and recycled six times without losing its catalytic activity. The results showed good yields of the Sonogashira products from the reactions with aryl iodides having EDGs and EWGs, regardless of the position of the substituent. However, reactions with aryl bromides gave lower yields leaving some unreacted starting materials.

A new approach to bimetallic Pd-Au/C nanoparticles (Scheme 46) for the Sonogashira reaction has been developed.¹⁵⁶

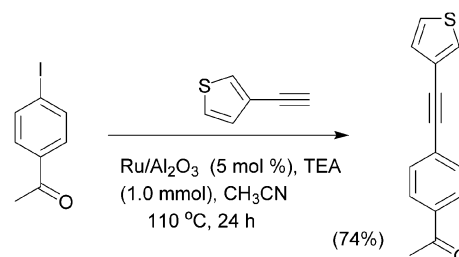


Scheme 46 Preparation of Pd-Au/C nano catalysts.

In this procedure Au atoms stabilize catalytically active Pd-NPs when engaged in an alloy heterogenized on carbon. The increased durability makes the Pd-Au/C catalyst more recyclable than the Au-free Pd/C catalyst. However, the substrate scope is limited to aryl iodides with different functional groups.

The reports in the literature applying Pd-NPs as catalysts can be regarded as an important step towards a simple system with the potential for commercial exploitation of the nanocatalysts. It has already been demonstrated that leached Pd(0) species from the catalyst surface are the true catalytic species in various cross coupling reactions catalyzed by Pd-NPs.^{157,158} Due to the operational simplicity and robustness of these nanocatalysts we envisage that they can be further developed to be applied in other Pd promoted transformations in the future.

3.3.2. Other transition metal nanoparticles as catalysts. As previously described, over the past decade, several research groups have been investigating the development of new Pd based nanocatalysts for the Sonogashira reaction. Toxicological features related to utilizing Pd catalysts have placed some limitations on their general use for large scale operations, specifically in the pharmaceutical industry. In 2004, a study demonstrated that Ru nanoparticle supported on alumina¹⁵⁹ could be used as a convenient heterogeneous catalyst system for the Sonogashira reaction in the absence of Cu co-catalyst which was influenced significantly by the nature and structure of metal supports (Scheme 47). The electronic nature of the alkynyl substituents has influence some impact on the reaction efficiency, however those with EDGs slightly retarded the reaction,



Scheme 47 Ru nanoparticle catalyzed Sonogashira reaction.

requiring higher temperatures. Satisfactory yields were obtained from the reactions of phenylacetylene with various aryl iodides regardless the electronic nature of electrophilic reagents.

The Sonogashira reaction has also been studied in the presence of metallic Rh nanoparticle¹⁶⁰ catalysts through surface-mediated heterogeneous catalysis. In this study the larger (8 nm) nanoparticles were found to be much better catalysts than small ones (2 nm), which is consistent with the hypothesis that steric limitations adversely affect the efficiency of the latter. However, the substrate scope is limited to aryl iodides. The paramagnetic magnetite (Fe_3O_4) nanoparticles (<30 nm)¹⁶¹ have been shown to be efficient as catalysts for the Sonogashira reaction under heterogeneous ligand-free conditions in ethylene glycol (Scheme 48). In this case, the catalyst can be easily separated by an external magnetic field and recycled for several consecutive runs without appreciable loss of its activity. But, the reaction of aryl bromides with terminal alkynes was disappointing and unreacted substrates were observed. However, it was also observed that activated aryl bromides such as 5-bromopyrimidine and 3-bromopyridine reacted smoothly in the presence of this catalyst and the desired alkynes were obtained in 76% and 80%, respectively.

It was reported that Ag nanocomposites¹⁶² can act as a catalyst in the Sonogashira reaction (Scheme 49). The activity of such material can be superior when compared to similar sized Pd nanocomposites, due to the unpaired electron of Ag in 5s orbital level which can easily be lost to form Ag^+ , which is beneficial to the key oxidative addition step in the Sonogashira reaction. This method exhibited good catalytic activity in Sonogashira reactions with different EWGs and EDGs connected to aryl halides, however in case of aryl chlorides only trace amount of coupling product was detected. Also, the Ag-

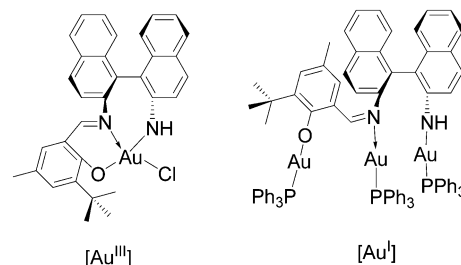


Fig. 8 Structures of different gold nano complexes.

NCs can be recycled and reused three times without loss of catalytic activity.

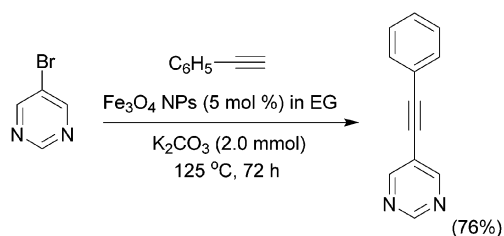
Currently, gold nanoparticles (Au-NPs) (Fig. 8) have received significant attention in the Sonogashira reactions due to their immense activity and selectivity. In 2007, a solid and stable Au supported on nanocrystalline CeO_2 (Au/CeO_2)⁵⁸ heterogeneous catalyst which is active for performing the Cu-free Sonogashira reaction has been developed. Although the catalyst has some significant advantages, including no Au leaching and good recyclability, it produces the homocoupled products.

Notably, in the same issue, another study reported that Au nanoparticles supported on CeO_2 or La_2O_3 ¹⁶³ catalysts exhibit strongly enhanced selectivity toward Sonogashira reactions. Very recently, it was reported that thiolate-protected $\text{Au}_{25}(\text{SCH}_2\text{-CH}_2\text{Ph})_{18}$ nanoclusters¹⁶⁴ supported on oxides (such as CeO_2 , TiO_2 , MgO , and SiO_2) able to catalyze Sonogashira reaction with high selectivity. DFT calculations of the co-adsorption of phenylacetylene and iodobenzene on the surface of the nanocluster found that the open facets of the nanocluster can strongly attract both reactants into a configuration ready for coupling. Research into the preparation and use of supported Au-NPs as a catalyst in processes such as the Sonogashira reaction has been reported in recent years, although in most cases only aryl iodides have been reported as coupling partners.

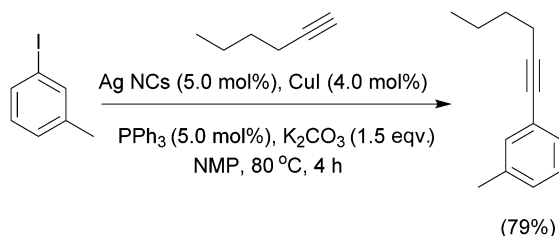
4. Outlook and summary

Among the newly developed catalytic systems, Pd-NPs possess a noteworthy position in the Sonogashira coupling reaction, although their catalytic activity and recyclability results are sometimes limited. In some cases, Cu, Fe, Ni, Ag or Au nanoparticles have been reported to act as reasonable catalysts in this protocol. Non-transition metal catalyzed and photo-induced Sonogashira reactions have opened a new window in the cross-coupling methodology. Although aryl iodides and bromides remain the most commonly used substrates, newer approaches, including supported nanoparticles and transition metal free systems can activate the more challenging aryl chlorides and aryl fluoride substrates.

Mechanistic studies have also provided useful information for the development and detailed understanding of conventional Sonogashira reactions. We envisage that in the near future new, more efficient and green protocols for Sonogashira reactions will become available, enabling the production of a



Scheme 48 Magnetite (Fe_3O_4) nanoparticles catalyzed Sonogashira reaction.



Scheme 49 Sonogashira reaction by using Ag-NPs as catalyst.

large array of compounds from the most diverse starting materials.

Acknowledgements

We are grateful to Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq) and Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES) for research fellowships (LCAB, MK), and Fundação de Amparo à Pesquisa de Minas Gerais (FAPEMIG), for financial support. GH thanks DIT for research support.

References

- 1 K. Sonogashira, Y. Tohda and N. Hagihara, *Tetrahedron Lett.*, 1975, **16**, 4467–4470.
- 2 M. Toyota, C. Komori and M. Ihara, *J. Org. Chem.*, 2000, **65**, 7110–7113.
- 3 I. I. Paterson, R. D. Davies and R. Marquez, *Angew. Chem., Int. Ed.*, 2001, **40**, 603–607.
- 4 N. D. Cosford, L. Tehrani, J. Roppe, E. Schweiger, N. D. Smith, J. Anderson, L. Bristow, J. Brodtkin, X. Jiang, I. McDonald, S. Rao, M. Washburn and M. A. Varney, *J. Med. Chem.*, 2003, **46**, 204–206.
- 5 M. C. Bagley, J. W. Dale, E. A. Merritt and X. Xiong, *Chem. Rev.*, 2005, **105**, 685–714.
- 6 L. Brunsveld, E. W. Meijer, R. B. Prince and J. S. Moore, *J. Am. Chem. Soc.*, 2001, **123**, 7978–7984.
- 7 M. Sonoda, A. Inaba, K. Itahashi and Y. Tobe, *Org. Lett.*, 2001, **3**, 2419–2421.
- 8 O. Mongin, L. Porres, L. Moreaux, J. Mertz and M. Blanchard-Desce, *Org. Lett.*, 2002, **4**, 719–722.
- 9 B. Li, Y. Fu, Y. Han and Z. Bo, *Macromol. Rapid Commun.*, 2006, **27**, 1355–1361.
- 10 D.-J. Bak, S.-C. Han, S.-H. Jin and J.-W. Lee, *Bull. Korean Chem. Soc.*, 2011, **32**, 3211–3212.
- 11 T. Hagiwara, Y. Murano, Y. Watanabe, T. Hoshi and T. Sawaguchi, *Tetrahedron Lett.*, 2012, **53**, 2805–2808.
- 12 S. Hoger, S. Rosselli, A. D. Ramminger and V. Enkelmann, *Org. Lett.*, 2002, **4**, 4269–4272.
- 13 Y. Tobe, N. Utsumi, A. Nagano and K. Naemura, *Angew. Chem., Int. Ed.*, 1998, **37**, 1285–1287.
- 14 K. Onitsuka, M. Fujimoto, N. Ohshiro and S. Takahashi, *Angew. Chem., Int. Ed.*, 1999, **38**, 689–692.
- 15 C. Glaser, *Ber. Dtsch. Chem. Ges.*, 1869, **2**, 422–424.
- 16 A. S. Hay, *J. Org. Chem.*, 1962, **27**, 3320–3321.
- 17 R. Chinchilla and C. Najera, *Chem. Rev.*, 2007, **107**, 874–922.
- 18 J. H. Li, J. L. Li, D. P. Wang, S. F. Pi, Y. X. Xie, M. B. Zhang and X. C. Hu, *J. Org. Chem.*, 2007, **72**, 2053–2057.
- 19 T. Fukuyama, M. Shinmen, S. Nishitani, M. Sato and I. Ryu, *Org. Lett.*, 2002, **4**, 1691–1694.
- 20 M. Beaupérin, E. Fayad, R. Amardeil, H. Cattey, P. Richard, S. Brandès, P. Meunier and J.-C. Hierro, *Organometallics*, 2008, **27**, 1506–1513.
- 21 J. Gil-Moltó and C. Nájera, *Eur. J. Org. Chem.*, 2005, **2005**, 4073–4081.
- 22 Y. Liang, Y. X. Xie and J. H. Li, *J. Org. Chem.*, 2006, **71**, 379–381.
- 23 K. G. Thakur, E. A. Jaseer, A. B. Naidu and G. Sekar, *Tetrahedron Lett.*, 2009, **50**, 2865–2869.
- 24 J. T. Guan, G.-A. Yu, L. Chen, T. Qing Weng, J. J. Yuan and S. H. Liu, *Appl. Organomet. Chem.*, 2009, **23**, 75–77.
- 25 M. Carril, A. Correa and C. Bolm, *Angew. Chem., Int. Ed.*, 2008, **47**, 4862–4865.
- 26 I. P. Beletskaya, G. V. Latyshev, A. V. Tsvetkov and N. V. Lukashev, *Tetrahedron Lett.*, 2003, **44**, 5011–5013.
- 27 P. Siemsen, R. C. Livingston and F. Diederich, *Angew. Chem., Int. Ed.*, 2000, **39**, 2632–2657.
- 28 D. Gelman and S. Buchwald, *Angew. Chem., Int. Ed.*, 2003, **42**, 5993–5996.
- 29 M. Aufiero, F. Proutiere and F. Schoenebeck, *Angew. Chem., Int. Ed.*, 2012, **51**, 7226–7230.
- 30 L. Sikk, J. Tammiku-Taul, P. Burk and A. Kotschy, *J. Mol. Model.*, 2012, **18**, 3025–3033.
- 31 M. García-Melchor, A. A. C. Braga, A. Lledós, G. Ujaque and F. Maseras, *Acc. Chem. Res.*, 2013, **46**, 2626–2634.
- 32 P. Burk, J. Tammiku-Taul and L. Sikk, *Proc. Est. Acad. Sci.*, 2013, **62**, 133–140.
- 33 P. V. D. Weghe, *Lett. Org. Chem.*, 2005, **2**, 113–117.
- 34 K. Sonogashira, in *Metal-Catalyzed Cross-Coupling Reactions*, Wiley-VCH, Weinheim, 2004.
- 35 A. L. Casado and P. Espinet, *Organometallics*, 1998, **17**, 954–959.
- 36 R. Á. O. N. Faza, A. R. de Lera and D. J. Cárdenas, *Adv. Synth. Catal.*, 2007, **349**, 887–906.
- 37 P. Fitton and E. A. Rick, *J. Organomet. Chem.*, 1971, **28**, 287–291.
- 38 P. Bertus, F. Fecourt, C. Bauder and P. Pale, *New J. Chem.*, 2004, **28**, 12–14.
- 39 C. He, J. Ke, H. Xu and A. Lei, *Angew. Chem., Int. Ed.*, 2013, **52**, 1527–1530.
- 40 L.-P. Chen, S.-G. Hong and H.-Q. Hou, *Chin. J. Struct. Chem.*, 2008, **27**, 1404–1411.
- 41 L. Sikk, J. Tammiku-Taul and P. Burk, *Organometallics*, 2011, **30**, 5656–5664.
- 42 M. García-Melchor, M. C. Pacheco, C. Nájera, A. Lledós and G. Ujaque, *ACS Catal.*, 2012, **2**, 135–144.
- 43 H. A. Dieck and F. R. Heck, *J. Organomet. Chem.*, 1975, **93**, 259–263.
- 44 C. Amatore, S. Bensalem, S. Ghalem and A. Jutand, *J. Organomet. Chem.*, 2004, **689**, 4642–4646.
- 45 A. Soheili, J. Albaneze-Walker, J. A. Murry, P. G. Dormer and D. L. Hughes, *Org. Lett.*, 2003, **5**, 4191–4194.
- 46 T. Ljungdahl, T. Bennur, A. Dallas, H. Emténäs and J. Mårtensson, *Organometallics*, 2008, **27**, 2490–2498.
- 47 C. Hansch, A. Leo and R. W. Taft, *Chem. Rev.*, 1991, **91**, 165–195.
- 48 S. Ehrenson, R. T. C. Brownlee and R. W. Taft, in *Progress in Physical Organic Chemistry*, John Wiley & Sons, Inc., 2007, pp. 1–80.
- 49 T. Ljungdahl, K. Pettersson, B. Albinsson and J. Mårtensson, *J. Org. Chem.*, 2006, **71**, 1677–1687.

- 50 K. H. Shaughnessy, P. Kim and J. F. Hartwig, *J. Am. Chem. Soc.*, 1999, **121**, 2123–2132.
- 51 A. Corma, R. Juarez, M. Boronat, F. Sanchez, M. Iglesias and H. Garcia, *Chem. Commun.*, 2011, **47**, 1446–1448.
- 52 F. Barrios-Landeros, B. P. Carrow and J. F. Hartwig, *J. Am. Chem. Soc.*, 2009, **131**, 8141–8154.
- 53 D. V. Partyka, M. Zeller, A. D. Hunter and T. G. Gray, *Angew. Chem., Int. Ed.*, 2006, **45**, 8188–8191.
- 54 A. S. Hashmi, C. Lothschütz, R. Dopp, M. Rudolph, T. D. Ramamurthi and F. Rominger, *Angew. Chem., Int. Ed.*, 2009, **48**, 8243–8246.
- 55 A. L. Casado and P. Espinet, *Organometallics*, 1998, **17**, 3677–3683.
- 56 T. Lauterbach, M. Livendahl, A. Rosellón, P. Espinet and A. M. Echavarren, *Org. Lett.*, 2010, **12**, 3006–3009.
- 57 B. Panda and T. K. Sarkar, *Chem. Commun.*, 2010, **46**, 3131–3133.
- 58 C. Gonzalez-Arellano, A. Abad, A. Corma, H. Garcia, M. Iglesias and F. Sanchez, *Angew. Chem., Int. Ed.*, 2007, **46**, 1536–1538.
- 59 C. Torborg and M. Beller, *Adv. Synth. Catal.*, 2009, **351**, 3027–3043.
- 60 I. P. Beletskaya and A. V. Cheprakov, *J. Organomet. Chem.*, 2004, **689**, 4055–4082.
- 61 C. A. Fleckenstein and H. Plenio, *Organometallics*, 2008, **27**, 3924–3932.
- 62 X.-Y. Chen, C. Barnes, J. R. Dias and T. C. Sandreczki, *Chem.–Eur. J.*, 2009, **15**, 2041–2044.
- 63 I. Thome, A. Nijs and C. Bolm, *Chem. Soc. Rev.*, 2012, **41**, 979–987.
- 64 J. Magano and J. Dunetz, *Chem. Rev.*, 2011, **111**, 2177–2250.
- 65 K. Heuze, D. Mery, D. Gauss and D. Astruc, *Chem. Commun.*, 2003, 2274–2275.
- 66 L. Yang, P. Guan, P. He, Q. Chen, C. Cao, Y. Peng, Z. Shi, G. Pang and Y. Shi, *Dalton Trans.*, 2012, **41**, 5020–5025.
- 67 P. M. Price, J. H. Clark and D. J. Macquarrie, *J. Chem. Soc., Dalton Trans.*, 2000, 101–110.
- 68 J. Zhu, J. Zhou, T. Zhao, X. Zhou, D. Chen and W. Yuan, *Appl. Catal., A*, 2009, **352**, 243–250.
- 69 K. K. R. Datta, M. Eswaramoorthy and C. N. R. Rao, *J. Mater. Chem.*, 2007, **17**, 613–615.
- 70 K. B. Sidhpuria, H. A. Patel, P. A. Parikh, P. Bahadur, H. C. Bajaj and R. V. Jasra, *Appl. Clay Sci.*, 2009, **42**, 386–390.
- 71 B. Karimi and D. Enders, *Org. Lett.*, 2006, **8**, 1237–1240.
- 72 M. Choi, D. H. Lee, K. Na, B. W. Yu and R. Ryoo, *Angew. Chem., Int. Ed.*, 2009, **48**, 3673–3676.
- 73 F. Z. Su, Y. M. Liu, L. C. Wang, Y. Cao, H. Y. He and K. N. Fan, *Angew. Chem., Int. Ed.*, 2008, **47**, 334–337.
- 74 A. Khalafi-Nezhad and F. Panahi, *Green Chem.*, 2011, **13**, 2408–2415.
- 75 N. Leadbeater, M. Marco and B. Tominack, *Org. Lett.*, 2003, **5**, 3919–3922.
- 76 P. Appukkuttan, W. Dehaen and E. Van der Eycken, *Eur. J. Org. Chem.*, 2003, **2003**, 4713–4716.
- 77 D. Prajapati, H. N. Borah and R. C. Boruah, *Synlett*, 2005, 2823–2825.
- 78 B. Prüger, G. Hofmeister, C. Jacobsen, D. Alberg, M. Nielsen and K. Jørgensen, *Chemistry*, 2010, **16**, 3783–3790.
- 79 M. Maji, S. Murarka and A. Studer, *Org. Lett.*, 2010, **12**, 3878–3881.
- 80 G. Jin, X. Zhang and S. Cao, *Org. Lett.*, 2013, **15**, 3114–3117.
- 81 R. Luque and D. Macquarrie, *Org. Biomol. Chem.*, 2009, **7**, 1627–1632.
- 82 T. Truong and O. Daugulis, *Org. Lett.*, 2011, **13**, 4172–4175.
- 83 R. K. Arvela, N. E. Leadbeater, M. S. Sangi, V. A. Williams, P. Granados and R. D. Singer, *J. Org. Chem.*, 2005, **70**, 161–168.
- 84 A. Sagadevan and K. C. Hwang, *Adv. Synth. Catal.*, 2012, **354**, 3421–3427.
- 85 S. Protti, M. Fagnoni, M. Mella and A. Albini, *J. Org. Chem.*, 2004, **69**, 3465–3473.
- 86 B. Guizzardi, M. Mella, M. Fagnoni and A. Albini, *J. Org. Chem.*, 2003, **68**, 1067–1074.
- 87 S. Milanese, M. Fagnoni and A. Albini, *J. Org. Chem.*, 2005, **70**, 603–610.
- 88 T. Okuyama, *Acc. Chem. Res.*, 2001, **35**, 12–18.
- 89 S. Protti, M. Fagnoni and A. Albini, *Angew. Chem., Int. Ed.*, 2005, **44**, 5675–5678.
- 90 M. Osawa, H. Nagai and M. Akita, *Dalton Trans.*, 2007, 827–829.
- 91 P. Schroll, D. P. Hari and B. König, *ChemistryOpen*, 2012, **1**, 130–133.
- 92 A. Fusano, T. Fukuyama, S. Nishitani, T. Inouye and I. Ryu, *Org. Lett.*, 2010, **12**, 2410–2413.
- 93 F. R. Lemke and C. P. Kubiak, *J. Organomet. Chem.*, 1989, **373**, 391–400.
- 94 H. Fischer, *Chem. Rev.*, 2001, **101**, 3581–3610.
- 95 K.-S. Focsaneanu, C. Aliaga and J. C. Scaiano, *Org. Lett.*, 2005, **7**, 4979–4982.
- 96 B. R. Buckley, S. E. Dann, D. P. Harris, H. Heaney and E. C. Stubbs, *Chem. Commun.*, 2010, **46**, 2274–2276.
- 97 A. R. Chaudhary and A. V. Bedekar, *Tetrahedron Lett.*, 2012, **53**, 6100–6103.
- 98 L. M. Rossi, N. J. S. Costa, J. Limberger and A. L. Monteiro, in *Nanocatalysis Synthesis and Applications*, John Wiley & Sons, Inc., 2013, pp. 51–88.
- 99 R. Chinchilla and C. Najera, *Chem. Soc. Rev.*, 2011, **40**, 5084–5121.
- 100 A. Balanta, C. Godard and C. Claver, *Chem. Soc. Rev.*, 2011, **40**, 4973–4985.
- 101 M. Moreno-Mañas and R. Pleixats, *Acc. Chem. Res.*, 2003, **36**, 638–643.
- 102 B. G. Johnson, *Top. Catal.*, 2003, **24**, 147–159.
- 103 R. Narayanan and M. A. El-Sayed, *Langmuir*, 2005, **21**, 2027–2033.
- 104 C. S. Consorti, F. R. Flores and J. Dupont, *J. Am. Chem. Soc.*, 2005, **127**, 12054–12065.
- 105 R. Chinchilla and C. Nájera, in *Nanocatalysis Synthesis and Applications*, John Wiley & Sons, Inc., 2013, pp. 89–131.
- 106 B. M. Choudary, S. Madhi, N. S. Chowdari, M. L. Kantam and B. Sreedhar, *J. Am. Chem. Soc.*, 2002, **124**, 14127–14136.
- 107 D. Astruc, K. Heuzé, S. Gatard, D. Méry, S. Nlate and L. Plault, *Adv. Synth. Catal.*, 2005, **347**, 329–338.

- 108 D. Astruc and V. Martinez, in *Metathesis Chemistry*, ed. Y. Imamoglu, V. Dragutan and S. Karabulut, Springer, Netherlands, 2007, vol. 243, pp. 223–236.
- 109 A. Roucoux, in *Surface and Interfacial Organometallic Chemistry and Catalysis*, ed. C. Copéret and B. Chaudret, Springer, Berlin Heidelberg, 2005, vol. 16, pp. 261–279.
- 110 C. Xue, K. Palaniappan, G. Arumugam, S. A. Hackney, J. Liu and H. Liu, *Catal. Lett.*, 2007, **116**, 94–100.
- 111 B. N. Lin, S. H. Huang, W. Y. Wu, C. Y. Mou and F. Y. Tsai, *Molecules*, 2010, **15**, 9157–9173.
- 112 T. V. Magdesieva, O. M. Nikitin, E. V. Zolotukhin, V. A. Zinovieva and M. A. Vorotyntsev, *Mendeleev Commun.*, 2012, **22**, 305–306.
- 113 Y.-T. Chu, K. Chanda, P.-H. Lin and M. H. Huang, *Langmuir*, 2012, **28**, 11258–11264.
- 114 A. R. Gholap, K. Venkatesan, R. Pasricha, T. Daniel, R. J. Lahoti and K. V. Srinivasan, *J. Org. Chem.*, 2005, **70**, 4869–4872.
- 115 P. Li, L. Wang and H. Li, *Tetrahedron*, 2005, **61**, 8633–8640.
- 116 B. M. Samarasimhareddy, G. Prabhu and V. V. Sureshbabu, *Tetrahedron Lett.*, 2014, **55**, 2256–2260.
- 117 M. Bakherad, A. Keivanloo and S. Samangoeei, *Chin. J. Catal.*, 2014, **35**, 324–328.
- 118 L. Chen, S. Hong, X. Zhou, Z. Zhou and H. Hou, *Catal. Commun.*, 2008, **9**, 2221–2225.
- 119 J. Athilakshmi, S. Ramanathan and D. K. Chand, *Tetrahedron Lett.*, 2008, **49**, 5286–5288.
- 120 S. Sawoo, D. Srimani, P. Dutta, R. Lahiri and A. Sarkar, *Tetrahedron*, 2009, **65**, 4367–4374.
- 121 S. Gao, N. Zhao, M. Shu and S. Che, *Appl. Catal., A*, 2010, **388**, 196–201.
- 122 N. T. S. Phan and H. V. Le, *J. Mol. Catal. A: Chem.*, 2011, **334**, 130–138.
- 123 M. Pagliaro, V. Pandarus, F. Beland, R. Ciriminna, G. Palmisano and P. D. Cara, *Catal. Sci. Technol.*, 2011, **1**, 736–739.
- 124 P. Li, L. Wang, L. Zhang and G.-W. Wang, *Adv. Synth. Catal.*, 2012, **354**, 1307–1318.
- 125 M. Esmaeilpour, A. R. Sardarian and J. Javidi, *J. Organomet. Chem.*, 2014, **749**, 233–240.
- 126 E. Mohsen, J. Jaber, M. A. Mehdi and N. D. Fatemeh, *J. Iran. Chem. Soc.*, 2013, **11**, 499–510.
- 127 M. Nasrollahzadeh, M. Maham, A. Ehsani and M. Khalaj, *RSC Adv.*, 2014, **4**, 19731.
- 128 H. Firouzabadi, N. Iranpoor, F. Kazemi and M. Gholinejad, *J. Mol. Catal. A: Chem.*, 2012, **357**, 154–161.
- 129 S. Moussa, A. R. Siamaki, B. F. Gupton and M. S. El-Shall, *ACS Catal.*, 2011, **2**, 145–154.
- 130 B. Tamami, M. M. Nezhad, S. Ghasemi and F. Farjadian, *J. Organomet. Chem.*, 2013, **743**, 10–16.
- 131 Y.-T. Hung, M.-T. Chen, M.-H. Huang, T.-Y. Kao, Y.-S. Liu and L.-C. Liang, *Inorg. Chem. Front.*, 2014, **1**, 405.
- 132 B. Tamami, F. Farjadian, S. Ghasemi and H. Allahyari, *New J. Chem.*, 2013, **37**, 2011–2018.
- 133 K. Niknam, A. Deris, F. Panahi and M. Reza Hormozi Nezhad, *J. Iran. Chem. Soc.*, 2013, **10**, 1291–1296.
- 134 B. J. Borah and D. K. Dutta, *J. Mol. Catal. A: Chem.*, 2013, **366**, 202–209.
- 135 K. H. Lee, S.-W. Han, K.-Y. Kwon and J. B. Park, *J. Colloid Interface Sci.*, 2013, **403**, 127–133.
- 136 A. Modak, J. Mondal and A. Bhaumik, *Green Chem.*, 2012, **14**, 2840–2855.
- 137 M. Navidi, N. Rezaei and B. Movassagh, *J. Organomet. Chem.*, 2013, **743**, 63–69.
- 138 D. Ganapathy and G. Sekar, *Catal. Commun.*, 2013, **39**, 50–54.
- 139 A. Khazaei, S. Rahmati and S. Saednia, *Catal. Commun.*, 2013, **37**, 9–13.
- 140 S. Keesara, S. Parvathaneni, G. Dussa and M. R. Mandapati, *J. Organomet. Chem.*, 2014, **765**, 31–38.
- 141 M. Hosseini-Sarvari, Z. Razmi and M. M. Doroodmand, *Appl. Catal., A*, 2014, **475**, 477–486.
- 142 M. Planellas, Y. Moglie, F. Alonso, M. Yus, R. Pleixats and A. Shafir, *Eur. J. Org. Chem.*, 2014, **2014**, 3001–3008.
- 143 S. Santra, K. Dhara, P. Ranjan, P. Bera, J. Dash and S. K. Mandal, *Green Chem.*, 2011, **13**, 3238–3247.
- 144 L. A. Aronica, A. M. Caporusso, G. Tuci, C. Evangelisti, M. Manzoli, M. Botavina and G. Martra, *Appl. Catal., A*, 2014, **480**, 1–9.
- 145 A. S. Roy, J. Mondal, B. Banerjee, P. Mondal, A. Bhaumik and S. M. Islam, *Appl. Catal., A*, 2014, **469**, 320–327.
- 146 S. U. Son, Y. Jang, J. Park, H. B. Na, H. M. Park, H. J. Yun, J. Lee and T. Hyeon, *J. Am. Chem. Soc.*, 2004, **126**, 5026–5027.
- 147 F. Bellina and M. Lessi, *Synlett*, 2012, **23**, 773–777.
- 148 P. Venkatesan and J. Santhanalakshmi, *Langmuir*, 2010, **26**, 12225–12229.
- 149 P. Cintas, G. Cravotto, E. C. Gaudino, L. Orio and L. Boffa, *Catal. Sci. Technol.*, 2012, **2**, 85–87.
- 150 M. Korzec, P. Bartczak, A. Niemczyk, J. Szade, M. Kapkowski, P. Zenderowska, K. Balin, J. Lełątko and J. Polanski, *J. Catal.*, 2014, **313**, 1–8.
- 151 D. Sengupta, J. Saha, G. De and B. Basu, *J. Mater. Chem. A*, 2014, **2**, 3986.
- 152 A. Shaabani and M. Mahyari, *J. Mater. Chem. A*, 2013, **1**, 9303–9311.
- 153 Y.-S. Feng, X.-Y. Lin, J. Hao and H.-J. Xu, *Tetrahedron*, 2014, **70**, 5249–5253.
- 154 J. Liu, X. Peng, W. Sun, Y. Zhao and C. Xia, *Org. Lett.*, 2008, **10**, 3933–3936.
- 155 J. Chung, J. Kim, Y. Jang, S. Byun, T. Hyeon and B. M. Kim, *Tetrahedron Lett.*, 2013, **54**, 5192–5196.
- 156 C. Rossy, J. Majimel, E. Fouquet, C. Delacôte, M. Boujtita, C. Labrugère, M. Tréguer-Delapierre and F.-X. Felpin, *Chem.-Eur. J.*, 2013, **19**, 14024–14029.
- 157 F. Zhao, B. M. Bhanage, M. Shirai and M. Arai, *Chem.-Eur. J.*, 2000, **6**, 843–848.
- 158 A. Biffis, M. Zecca and M. Basato, *Eur. J. Inorg. Chem.*, 2001, **2001**, 1131–1133.
- 159 S. Park, M. Kim, D. H. Koo and S. Chang, *Adv. Synth. Catal.*, 2004, **346**, 1638–1640.

- 160 V. K. Kanuru, S. M. Humphrey, J. M. W. Kyffin, D. A. Jefferson, J. W. Burton, M. Armbruster and R. M. Lambert, *Dalton Trans.*, 2009, 7602–7605.
- 161 H. Firouzabadi, N. Iranpoor, M. Gholinejad and J. Hoseini, *Adv. Synth. Catal.*, 2011, 353, 125–132.
- 162 M. Han, S. Liu, X. Nie, D. Yuan, P. Sun, Z. Dai and J. Bao, *RSC Adv.*, 2012, 2, 6061–6067.
- 163 S. K. Beaumont, G. Kyriakou and R. M. Lambert, *J. Am. Chem. Soc.*, 2010, 132, 12246–12248.
- 164 G. Li, D.-E. Jiang, C. Liu, C. Yu and R. Jin, *J. Catal.*, 2013, 306, 177–183.